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Chapter 2: Literature Review

Chapter 2: Literature Review

This chapter introduces theoretical and empirical literature review to be used in this study. The chapter begins with the introduction of regime-switching models used in finance, i.e., Markov-Switching Model (MSM) and Hidden Markov Model (HMM). The subsequent explanation places HMMs in the broader family of Probabilistic Graphical Models (PGMs) and proceeds with the introduction of an overview of how they are used to predict oil prices. The chapter ends with an enumeration of common gaps and limitations this study is attempting to bridge. 2.1 Regime-Switching Models in Finance

Financial markets show complex, non-linear behavior with regime switching and structural breaks, i.e., bull-bear or high-low volatility regimes. Strict linear time series models are not appropriate for describing such behavior. Regime-switching models address the above deficiency by positing model parameters to be controlled by a discrete-state stochastic process (Hamilton, 1994). Regime-switching models do therefore have a general form to estimate market state transition and investor sentiment regimes change.

2.1.1 Markov-Switching Models

Hamilton (1989) seminal paper made MS models popular in economics for regime-switching autoregressive (AR) model. Regime switching in MS models is governed by a first-order Markov chain, i.e., the regime in time t–1 alone affects the probability of changing the regime at time t. Since these regimes are usually unobservable (latent), they need to be estimated from the observed data.

A standard Markov-Switching Autoregressive (MS-AR) model is defined as:

$$y_t = \mu_{S_t} + \sum_{i=1}^{p} \phi_{i,S_t}(y_{t-i} - \mu_{S_{t-i}}) + \sigma_{S_t}\varepsilon_t$$

where S_t is an unobservable regime variable, and μ_{S_t}, ϕ_{i,S_t}, and σ_{S_t} are the regime-specific mean, autoregressive coefficients, and volatility, respectively. The error term εt is a white noise process (Hamilton, 1994).

The most significant individual advantage of MS models is that they can explain parsimoniously asymmetric cycles as well as long-lasting time-varying volatility. It is best applied to model business cycles, asset returns, and financial crises (Guidolin & Timmermann, 2007).

2.1.2 Hidden Markov Models (HMMs)

Hidden Markov Models (HMMs) are a generalization of the MS model by adding a probabilistic relationship between observable variables and hidden states. Although both MS and HMMs have been applied interchangeably in finance, HMMs better describe systems where observations are a result of a Markov process on hidden states.

2.1.2.1 Formal Definition of HMMs

- A discrete HMM is defined formally by Rabiner (1989) as follows:
- N: Number of hidden states, S={S1,S2,…,SN}, where q_t is the state at time t.
- M: Number of observation symbols or outputs. In finance applications, these are typically continuous (e.g., asset return), and they are described by probability density functions.
- A: State transition probability matrix, A=[aij]], where $a_{ij} = P(q_{t+1} = S_j | q_t = S_i)$.
- B: Observation probability distribution. In continuous HMMs, B is frequently modeled as a Gaussian mixture model:

$$b_j(o) = \sum_{m=1}^{M} c_{jm} N(o|\mu_{jm}, \Sigma_{jm})$$

where c_{jm} is the weight of the m-th component of state j.

π: Initial state distribution, π={πi}, where $\pi_i = P(q_1 = S_i)$.

The three basic problems of HMMs are:

- **Evaluation** – Finding the probability of an observation sequence given model parameters.
- **Decoding** –Finding the most likely sequence of hidden states.
- **Learning** –Finding model parameters λ=(A,B,π) to achieve maximum probability of observations (Rabiner, 1989).

2.1.2.2 Finance Applications

HMMs have been utilized successfully in a wide variety of finance applications since they have the capability to capture regime-dependent, dynamic behavior:

Asset Allocation and Return Modeling: Guidolin and Timmermann (2007) employed multivariate MS models to detect bull, bear, and crash regimes in asset returns and demonstrated substantial asset allocation strategy gains when regime effects are considered.

- **Risk Management and Volatility Clustering:** HMMs explain volatility clustering by associating high-volatility episodes with unique hidden states. This facilitates dynamic VaR and Conditional VaR estimation that is responsive to prevailing market conditions (Marriott & Pope, 2023).
- **Credit Rating Migration:** HMMs explain corporate credit rating movements most accurately through the utilization of the hidden state as a firm's underlying financial state, and credit ratings as observable as emissions (Bharath & Shumway, 2008).

2.2 Probabilistic Graphical Models

Probabilistic Graphical Models (PGMs) utilize the use of probability and graph theory in modeling sophisticated patterns of dependency between random variables (Koller & Friedman, 2009). PGMs provide the possibility for both the graphical and mathematical approach of reasoning with uncertainty and conditional dependency.

HMMs are a PGM, i.e., a Dynamic Bayesian Network (DBN). Hidden states qtq_tqt inside the model form a chain such that q_t is a function of q_{t-1}, alone, and for any observation oto_tot, it is a function of the corresponding state alone. These words impose two strong conditional independence assumptions on:

- For q_t, observation o_t t is a function of all variables but the other one.
- With q_{t-1}, state q_t is state-independent.

Probabilistic graphical model (PGM) formulation of an HMM makes it possible for researchers to parameterize and generalize the model, factorize a set of observation sequences, and take advantage of high-performance algorithms at inference to estimate parameters and learn (Koller & Friedman, 2009).

2.3 HMMs for Forecasting Oil Price

Forecasting of oil price is not a simple task because it is geopolitically sensitive, supply-demand disequilibrium prone, and prone to manipulations causing regime changes. It has been used exhaustively to forecast such complex dynamics through non-linear methods such as HMMs.

Previous studies, e.g., Bhar and Hamori (2004), showed the potential of HMMs to distinguish between high- and low-volatility regimes of oil price returns. Later studies focused on multi-state models that distinguish between various regimes like "stable growth," "price crash," and "price spike" (Janczura & Weron, 2010).

The process of forecasting is typically two-stepped:

- Estimation of the most probable sequence of the hidden states with the assistance of filtering algorithms.
- Utilizing regime-dependent parameters—regime-specific means and variances—when generating predictions.

Evidence on regime-switching models being more superior to conventional linear or random-walk models, particularly where regime persistence is appropriate for the behavior of prices in medium- and long-horizon forecasting (Nomikos & Pouliasis, 2015).

2.4 Deficiencies and Weaknesses of the Study

Although HMMs can model finance and oil price forecasting, there are some deficiencies so far:

Features Omission: Most studies utilize single-variable models that take into account only historical prices (Bhar & Hamori, 2004; Janczura & Weron, 2010), and not cross-market and fundamental drivers—equity indices, exchange rates, and production level—potentially useful for regime detection.

- **Model Specification Issues:** N selection is usually informal on the basis of AIC or BIC considerations rather than deliberative. There is no hard data-driven way to estimate state complexity.
- **Restricted Application of Higher-Frequency and Alternative Data Sources:** Some restricted applications of alternative or higher-frequency data sources (e.g., news sentiment, satellite images of warehouses) would significantly alter the accuracy of forecasts.
- **Interpretability of Regimes:** The states are post-hoc acquired in the majority of the studies ruling out economic fundamentals, and interpretability is unknown.

This project tries to overcome such shortcomings by formulating a multivariate Hidden Markov Model combining fundamentals and financial variables for regime detection enhancement and enhancing crude oil price forecasting.

Step 1

Forward and Backward Algorithms

Forward-backward algorithm is the core algorithm of inference and analysis of the Hidden Markov Model (HMM). The forward algorithm and the backward algorithm are two dual dynamic programming algorithms.

The forward algorithm calculates the probability of an observation sequence until a certain time and of being in a certain state at that time. It calculates this by recursively propagating probabilities through the state sequence. It calculates, in fact, the so-called "alpha" variables, that is, the joint probability of the observation sequence until time t and of the state at time t (Rabiner, 1989). This forward pass totals the evidence from the beginning of the sequence.

On the other hand, the backward algorithm goes in the opposite direction. It computes the probability of the next sequence of observations when one is currently in a certain state at time t. These are the "beta" variables, which produce the probability of the observations from time $t+1$ to the end point based on the state at time t (Rabiner, 1989). In the case of the forward algorithm, it consumes the "past" from a position, the backward algorithm consumes the "future."

Both of these algorithms resolve the intrinsic likelihood problem of HMMs, i.e., calculate the probability of the observed sequence under the model parameters (Rabiner, 1989). Additionally, with the cumulative alpha and beta variables, posterior probabilities of being in any state at any time can be calculated, which is a critical application in finding a solution to the decoding and learning problem of HMMs (Murphy, 2012).

Numerical underflow will likely be the biggest practical limitation to performing these algorithms. Since probabilities are being multiplied tens of thousands of times through tens of thousands of long sequences, alpha and beta's values build up extremely fast to extremely large figures outside the numerical range of digital computers, effectively zero. This is balanced by introducing scale factors at every step in order to maintain probabilities in some appropriate numerical range without losing essential likelihood ratios (Rabiner, 1989; Bishop, 2006). Scaling is necessary in order to make HMMs viable to apply on real data sequences.

Forward Algorithm Pseudocode

Input: Observations O[1..T], states S[1..N], transition A, emission B, initial distribution π
Initialize:

```
 for i = 1..N:
   α[1,i] = π[i] * B[i,O1]
```

Induction:

```
for t = 2..T:
  for j = 1..N:
    α[t,j] = (Σ_i α[t-1,i] * A[i,j]) * B[j,Ot]
```

Termination:

```
P(O|model) = Σ_j α[T,j]
```

Pseudocode (Backward Algorithm)

Initialize:

```
for i = 1..N:
  β[T,i] = 1
```

Induction:

```
for t = T-1 down to 1:
  for i = 1..N:
    β[t,i] = Σ_j (A[i,j] * B[j,O(t+1)] * β[t+1,j])
```

Termination:

```
P(O|model) = Σ_i π[i] * B[i,O1] * β[1,i]
```

Toy Example

Suppose we have two hidden states: Bull (S1), Bear (S2), and three observations: Up, Flat, Down.

Parameters:

- π = [0.5, 0.5]
- A = [[0.8, 0.2], [0.3, 0.7]]
- B = [[0.6, 0.2, 0.2], [0.2, 0.3, 0.5]]
  Observation: O = [Up, Flat]

Forward:

- α[1, Bull] = 0.5 * 0.6 = 0.30
- α[1, Bear] = 0.5 * 0.2 = 0.10
- α[2, Bull] = (0.30*0.8 + 0.10*0.3)*0.2 = 0.054
- α[2, Bear] = (0.30*0.2 + 0.10*0.7)*0.3 = 0.033
  Final likelihood = 0.087

Backward:

- β[2,*] = 1

- β[1,Bull] = 0.8*0.2*1 + 0.2*0.3*1 = 0.22
- β[1,Bear] = 0.3*0.2*1 + 0.7*0.3*1 = 0.27

  Likelihood = 0.5*0.6*0.22 + 0.5*0.2*0.27 = 0.087

Viterbi Algorithm Explanation

Viterbi algorithm is a dynamic program that is able to calculate the solution of the simple problem in Hidden Markov Models (HMMs): the decoding problem. It is an approximation of the most probable single sequence of hidden states (usually called the "Viterbi path") which would have generated a given sequence of observations (Viterbi, 1967; Forney, 1973).

Functionally, the Viterbi algorithm is essentially the same as the forward algorithm in the way that it handles observation sequence left to right. There remains an extremely vast disparity in its basic operation, however. While the forward algorithm is constructing probabilities for all possible paths so afterward it can estimate overall likelihood, the Viterbi algorithm focuses on finding best or highest-probability path. With every step, for every state, it calculates the highest probability of any path to the state and keeps a "traceback" pointer to record the optimal path (Rabiner, 1989). By repeatedly choosing the most likely predecessor, the global optimal path can be obtained without having to search all possible lists of states.

One of the most important uses of the Viterbi algorithm is the identification of regimes in financial markets. Asset prices, for example, are observable, but unobservable market regimes (bull and bear markets, high-volatility regimes) are not. One could then apply the Viterbi algorithm to a price movement-trained HMM to "decode" the most probable sequence of such hidden regimes that would be extremely helpful for risk management and quantitative strategy (Bulla & Bulla, 2006).

Pseudocode (Viterbi Algorithm)

```
Initialize:
  for i = 1..N:
    δ[1,i] = log(π[i]) + log(B[i,O1])
    ψ[1,i] = 0
Recursion:
  for t = 2..T:
    for j = 1..N:
      δ[t,j] = max_i (δ[t-1,i] + log(A[i,j])) + log(B[j,Ot])
      ψ[t,j] = argmax_i (δ[t-1,i] + log(A[i,j]))
```

Termination:

state_T = argmax_i δ[T,i]

Path backtracking:

for t = T-1 down to 1:

state_t = ψ[t+1, state_{t+1}]

Toy Example

Using the same parameters, with O = [Up, Flat, Down]:

- At t=1:
  δ[1,Bull] = log(0.5*0.6) = log(0.30)*
  *δ[1,Bear] = log(0.5*0.2) = log(0.10)

- At t=2, choose max paths for each state.

- At t=3, continue recursion.

Baum-Welch Algorithm Explanation

The Baum-Welch algorithm is an answer to the third elementary problem of Hidden Markov Models (HMMs) called the learning problem. In case model parameters—transition probabilities, emission probabilities, and initial state distribution—are unknown, the Baum-Welch algorithm will learn them out of an observation sequence database (Baum et al., 1970).

Baum-Welch is by theory an implementation of the Expectation-Maximization (EM) algorithm (Dempster, Laird, & Rubin, 1977). Consistent with it, it iteratively executes a two-step algorithm that will converge to a local likelihood function maximum. During the Expectation (E) step, it computes the expected hidden state and state transition occupancies given the prevailing model parameters. This is done using the forward and backward probabilities, where one finds out the probability of being in any state at any time and the probability of moving from one state to another, taking the entire observed sequence into account (Rabiner, 1989).

Finally, in the Maximization (M) step, the estimated expected counts are used as sufficient statistics to re-estimate HMM parameters. The transition matrix is maximized by the expected number of transitions through states, and the emission matrix is maximized by the expected number of times an observation had been emitted from a state (Murphy, 2012). This expectation-maximization iteration is repeated until the data probability under the model improves by less than some value, i.e., until convergence.

One of the key characteristics of this algorithm is that in each iteration, the model probability will increase or stay unchanged but never go down (Rabiner, 1989). This makes the Baum-Welch algorithm a stable and traditional way of unsupervised HMM topology estimation from sequence data.

Pseudocode (Baum-Welch Algorithm)

```
Initialize π, A, B randomly
Repeat until convergence:
  # E-step
  Compute α, β using forward and backward algorithms
  Compute γ[t,i] = P(state_t = i | O)
  Compute ξ[t,i,j] = P(state_t=i, state_{t+1}=j | O)

# M-step
  π[i] = γ[1,i]
  A[i,j] = Σ_{t=1 to T-1} ξ[t,i,j] / Σ_{t=1 to T-1} γ[t,i]
  B[i,k] = Σ_{t where O[t]=k} γ[t,i] / Σ_{t=1 to T} γ[t,i]
Until log-likelihood change < ε
```

Toy Example

Given the same model with unknown A and B, initialize random probabilities. Using O = [Up, Flat, Down], run one iteration:

- E-step: Forward-Backward yields γ and ξ.
- M-step: Update A and B. For example, A[1,1] might increase if γ suggests Bull persists, while B[2,Down] increases if Bear is strongly associated with Down.

Step 2

Bull Regimes Definition and

Characteristics

Bull regime in the market is a series of continuous optimism and resilience driven by humongous-scale trends of a direction of rising trend of asset prices, favorable mean returns, and overall decreased relative volatility compared to rough bear regimes (Chen, 2017). Numerically not only but actually supported by good economic times such as strong economic growth, upbeat investor sentiment, and high liquidity in commodity and equity markets (Alvi, 2018).

The style of bull personality regime is subsequently firmly established within the macroeconomy. Stock prices are projected economic good fortune sought-after equity returns, and therefore bull markets would subsequently be in line with increasing GDP, deflated inflation, and overall leading stock index tendencies, as Fama (1990) had predicted. Plotting a few key markers like S&P 500 Index (SP500_Index) and US Gross Domestic Product (US_GDP) shows the steadily upward trends in phases, which is an indication of positive expansionary forces. For example, sustained rises of the SP500_Index coupled with stable or rising GDP and well-managed CPI are indications of super-high corporate productivity culture along with all-time record consumer confidence levels. Besides that, an increase in long-term interest rates, i.e., as indicated by, e.g., the US 10-Year Treasury Yield (US10Y_Yield), will be a message that investors expect a higher future return and are shedding riskless assets in a "risk-on" message.

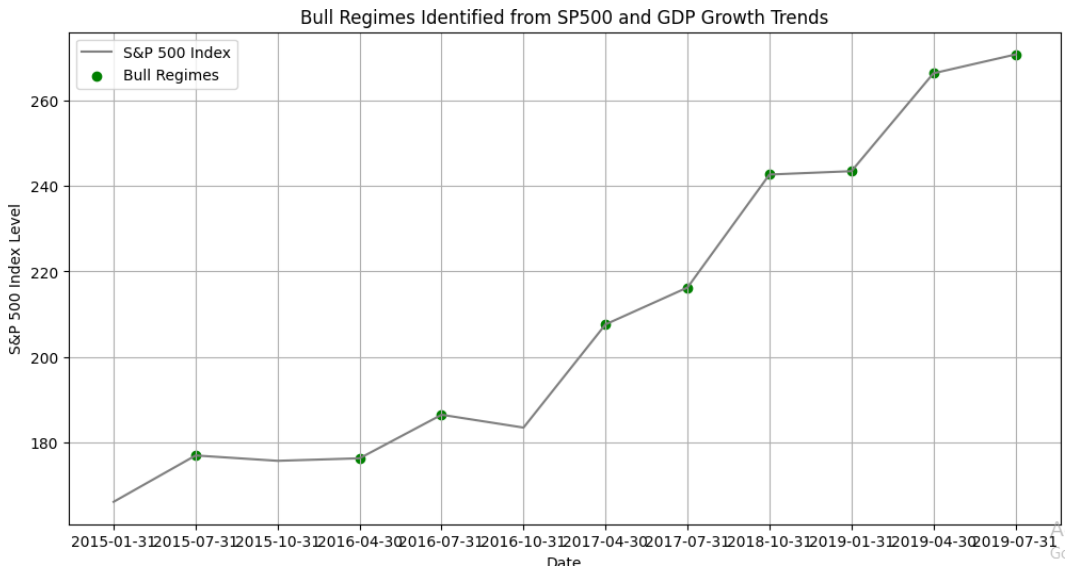
Bull power also finds voice in cross-asset trends. The US Dollar Index (USD_Index) increase is a reflection of money flowing into US assets, and it fortifies the self-reinforcing positive feedback loop. This overall bullness also manifests itself in more higher stock prices in big brushstrokes of industries from blue-shield energy giants like ExxonMobil (XOM) to growth stocks like Tesla (TSLA), and this renders bull sentiment indiscriminate and non-sector-specific.

Summary of Dominant Economic Features of Bull Regimes:

- **Macroeconomic Performance:** Rising GDP and relatively stable-to-rising moderately CPI, illustrating rising economic activity and consumption by consumers.
- **Market Performance:** Rising long-term trends of rising equity indices like the S&P 500 with cross-section participation.
- **Monetary and Credit Environment:** Rising moderately long-term bond yields, in line with projections of future inflation and growth.
- **International Confidence:** Rising USD_Index, usually a gauge of confidence in the world in the U.S. economy.

- **Sectoral Strength:** Difference in increase in stock price across sectors, continuing the market breadth up trend.

Figure 1: Plot showing Bull Regimes Identified from SP500 and GDP Growth Trends



Bear Regimes Definition and

Characteristics

Bear regime, or contraction regime, in financial markets refers to a well-characterized regime that is found with chronic and often deep declines in asset prices, negative mean returns, and record highs in market volatility. Bear regimes are usually continuous financial events but necessarily tied to contractions of the overall economy and typically occur together with recessions, financial crises, or ginormous geopolitical shockwaves that break investor confidence (Hamilton, 1989).

Fundamentally, bear markets consist of an interplay of bad economic fortune and extremely bearish investor sentiment. As Kim and Nelson (1999) deduce, they always entail shrinking economic activity (GDP), declining corporate profits and stock prices, and flights to safety lowering long-term bond yields. Financial time series data display uniformly clear-cut evidence of these forces. For instance, in the case of bear regimes, the ideal indicators like US Gross Domestic Product (US_GDP), S&P 500 Index (SP500_Index), and Crude Oil (Crude_Oil_WTI) prices will always present negative or poor trends.

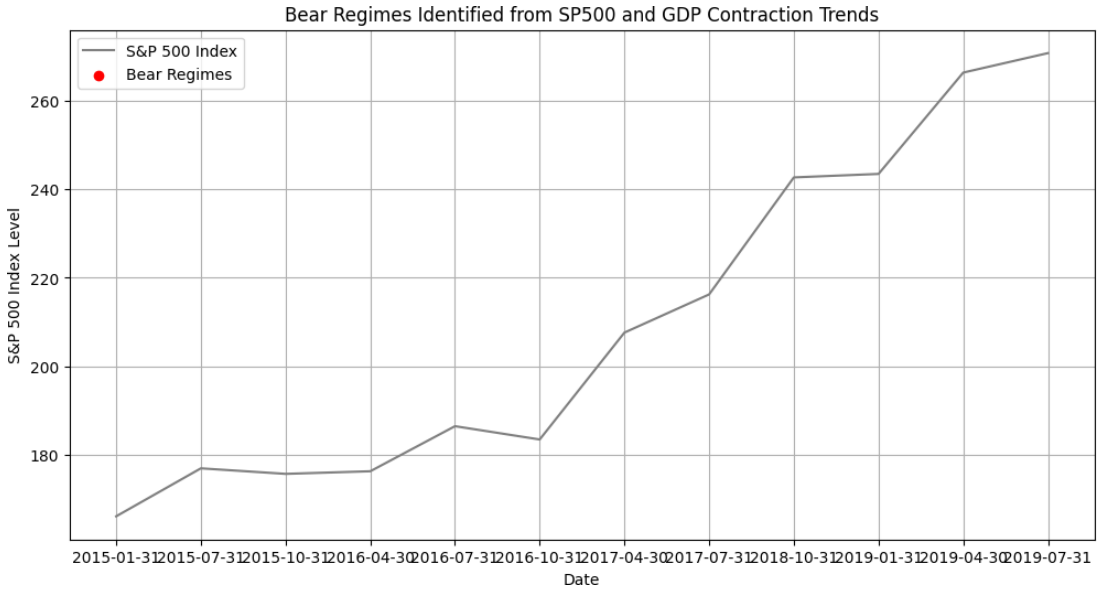
This economic downturn results in the classic "flight-to-safety" scenario. As investors shed risky positions like equities, they shift towards quantifiable safe-haven assets. The same effect is observed organically from the statistics through two classic adjustments: rising prices of US Treasury Bonds (which is inversely correlated to

falling yields) and ongoing appreciation of the US Dollar Index (USD_Index). The latter is a sign of a rush on a global level for comparative safety and liquidity in foreign distress in terms of dollars, producing crisis-capital flows.

Summary of Peak Economic Characteristics of Bear Regimes:

- Macroeconomic Decline: Falling GDP and slowing industrial demand, which implies a shrinking economy and lower aggregate demand.
- Market Decline: Sustained falls in broad equity indices e.g., S&P 500 due to falling company earnings and bearishness among investors.
- Flight to Quality: Rising government bond yields (i.e., US Treasuries) as investors prioritize the protection of capital over returns, leading to falling yields.
- Commodity Weakness: Disinflation of prices of major industrial commodities like oil, indicative of a sharp decline in economic and industrial activity worldwide.
- Risk Aversion: As a result of market tension, this is usually indicated by a strong US dollar which is an indicator of a global flight to risk.

Figure 2: Bear Regime Identified from SP500 and GDP Contraction Trends



Stagnant Regimes Definition and

Features

A stagnant, or sideways, regime market is a specific period of time in which there is an extremely evident lack of directional momentum, and the prices on assets fluctuate in an exceedingly narrow range for an unbelievably extended period of time. This causes the returns to hover around zero and is typically followed by low absolute market volatility. These phases illustrate a state of equilibrium or indecision, usually due to market uncertainty, thin trading volumes, or precarious balance between conflicting bullish and bearish pressures (Lo & MacKinlay, 1999).

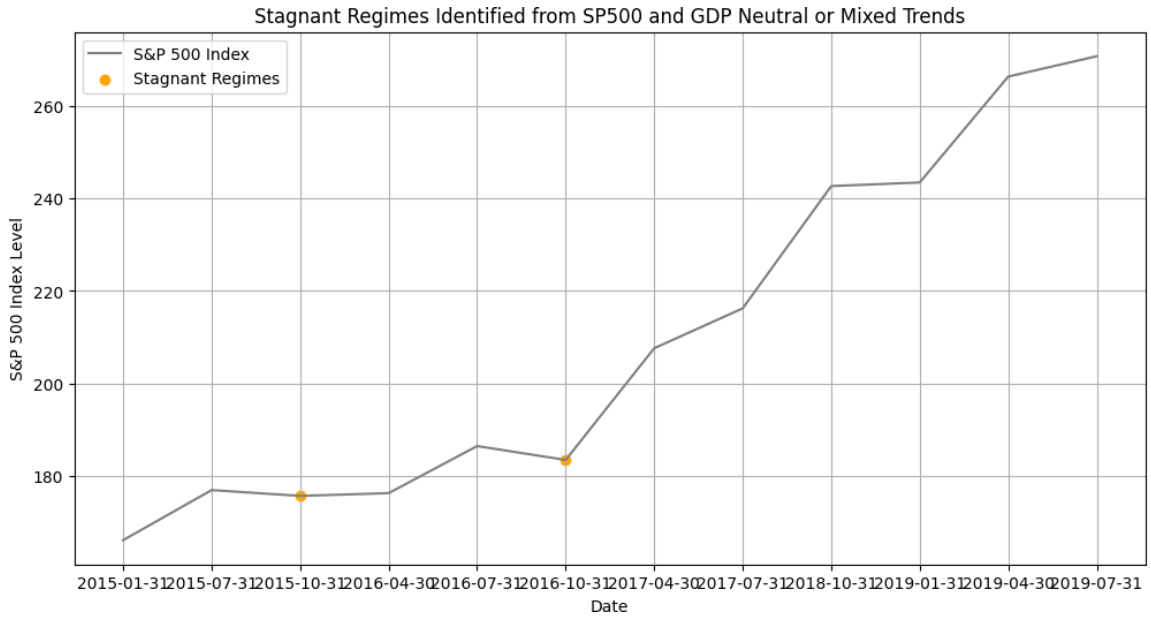
In addition to raw price action, stagnant regimes would also encompass the phase of structural adjustment, policy uncertainty, or market consolidation where financial markets and the economy take their time to adapt to previous gains or losses before aligning to a new trend (Hamilton, 2016). Tracking of major financial indicators easily detects such phases. In stagnations, the key benchmarks like the S&P 500 Index (SP500_Index), Crude Oil prices (Crude_Oil_WTI), and the US Dollar Index (USD_Index) barely fluctuate around mean levels and provide a widely accepted range for trading.

This kind of price action is associated with a macroeconomic standstill environment. The combination of tight Consumer Price Index (CPI) readings and flat Treasury yields is the message of a period of policy calm and the lack of fresh, paradigm-shattering economic data to drive monster buying or selling. That has them stumped, with offsetting macroeconomic forces—bullish growth indicators versus stubborn structural ills—pretty much neutralizing one another. The following balance in low volatility across a wide range of individual stocks of mature energy majors like Chevron (CVX) to cyclicals like Delta Air Lines (DAL) implies the market-wide rather than sector-wide.

Summary of Most Important Economic Features of Stagnant Regimes:

- **Market Consolidation:** Range trading or level trading within broad equity indices and commodity prices that indicate lack of directional conviction.
- **Macroeconomic Stability:** This indicates cross-over CPI and bond yields at attaining a policy stability with no inflationary or deflationary shock.
- **Low Volatility:** It is a depiction of the absence of ruggedness in stock and asset classes, indicating symmetry among market participants and absence of trading.
- **Investor Uncertainty:** This is the state of equilibrium in the market where both the bull story and the bear story are credibly possible and end up being money in hand and waiting-and-seeing.

Figure 3: Stagnant Regimes Identified from SP500 and GDP Neutral or Mixed Trends



Step 3

Definition of Hidden Markov Model

A Hidden Markov Model (HMM) is a strong statistical technique for modeling sequential data if the observed system should be a Markov process with hidden states. The basic concept is a two-level stochastic process: a hidden process that changes with time as a Markov chain, and an observable process depending on the underlying hidden state (Rabiner, 1989). We do not observe the sequence of states the system passes through but instead the emissions (measurements) stochastically produced by each state. This makes HMMs especially well adapted for use on time-series data where the driving forces cannot be observed, i.e., speech recognition, genomic sequence analysis, and regime detection in finance (Eddy, 2004).

The potentiality of the entire HMM structure is defined by three (3) fundamental assumptions that make computation and inference easy:

1. The Markov Property (State Transition):
The hidden state Markov chain is a first-order Markov chain. That is, the probability of transitioning to any specific hidden state at time t depends only on the current hidden state at time $t-1$, rather than on the entire past history of all states (Rabiner, 1989). Symbolically, it is expressed as $P(S_t|S_1, S_2, \ldots, S_{t-1}) = P(S_t|S_{t-1})$, where S_t represents the hidden state at time t. This "memorylessness" of the HMM is a simplifying assumption which allows the model to be computationally efficient with long sequences.

2. Conditional Independence of Observations (Emission Probability):
Every one of the observations in the sequence is conditionally independent of all the other observations given the current hidden state. Every one of the observations in the sequence is conditionally independent of all the other observations given the current hidden state. O_t depends only on the current hidden state, S_t, and not on past states or observations (Bishop, 2006). Mathematically., $P(O_t|S_1, \ldots, S_t, O_1, \ldots, O_{t-1}) = P(O_t|S_t)$. This assumption states that all in terms of the distribution of an observation is in the hidden state, and the observation does not give any additional information regarding future or past states or observations assuming the state is already known.

3. Time-Homogeneity (Stationarity):
We are assuming that transition probabilities among hidden states and emission probabilities of observations from states are stationary and invariant during the entire sequence (Rabiner, 1989). This means that the probability of transitioning from state to state j $(P(S_t = j|S_{t-1} = i))$ is independent of the time step t, and also the observation probability of a state given (P(O_t |S_t)) is independent of time. Although this is a standard

assumption, model specification has in recent releases been generalized to include time-varying parameters in order to accommodate non-stationarity in environments (Zucchini, MacDonald, & Langrock, 2016).

Formally, we define an HMM as a quintuple:

$$\lambda = (S,\ \pi,\ A,\ \mathcal{O},\ B)$$

where:

- $S = \{s_1, s_2, \ldots, s_N\}$ is the finite set of hidden states (e.g. “bull”, “bear”, “stagnant”).
- $\pi = \{\pi_i\}_{i=1}^{N}$ is the initial state distribution, where $\pi_i = P(s_1 = s_i)$.
- $A = [a_{ij}]$ is the transition probability matrix of size $N \times N$.

$$a_{ij} = P(s_t = s_j \mid s_{t-1} = s_i), \quad \sum_{j=1}^{N} a_{ij} = 1.$$

- $\mathcal{O} = \{o_1, o_2, \ldots, o_M\}$ is the set (alphabet) of possible observations (if discrete) or a suitable continuous observation space.
- $B = \{b_j(o)\}$ is the emission (or observation) probability model: $b_j(o) = P(o_t = o \mid s_t = s_j)$.

Under this model, the **joint probability** of a particular hidden-state sequence $s = (s_1, s_2, \ldots, s_T)$ and observed sequence $o = (o_1, o_2, \ldots, o_T)$ is:

$$P(\mathbf{s}, \mathbf{o} \mid \lambda) = \pi_{s_1} \cdot b_{s_1}(o_1) \cdot \prod_{t=2}^{T} \left[a_{s_{t-1},\, s_t} \cdot b_{s_t}(o_t)\right].$$

From this formulation arise the three canonical problems of HMMs:

1. **Evaluation / Likelihood**: compute $P(\mathbf{o} \mid \lambda)$ (via Forward/Backward algorithms).

2. **Decoding**: find the most likely hidden-state sequence. $\hat{\mathbf{s}} = \arg\max_{\mathbf{s}} P(\mathbf{s} \mid \mathbf{o}, \lambda)$
3. **Learning / Parameter estimation**: given observations ooo, estimate the model parameters $\lambda = (A, B, \pi)$ (via Baum–Welch / EM, or modern variants).

Contemporary perspectives & remarks

- While the classical HMM assumes discrete states and either discrete or Gaussian emissions, modern variants allow more flexible emission distributions (e.g. mixture models, autoregressive observations) or *nonstationary* transitions (time-varying A_t) (see Hsu, Kakade & Zhang 2008).
- HMMs can be interpreted as a simple form of **Dynamic Bayesian Network** (DBN) or a two-layer graphical model: a chain of latent nodes connected by transitions, each emitting an observable node at each time (Jurafsky & Martin, 2025).
- Some recent work critiques certain classical definitions (e.g. about model identifiability or hidden degeneracy) and proposes more rigorous formalizations of HMM variants (e.g. “On the definitions of hidden Markov models”) (see ScienceDirect review).

Step 4

Macroeconomic econometrics and economic forecasting form the pillars to unraveling enigmas behind complex market activity patterns. Regime detection in HMM largely relies, in a wonderful way, on the identification of appropriate economic surrogates, homogeneous and homogeneous sets of data derivation, and proper data cleaning. Regime-switching behavior—i.e., those easily recognizable bull, bear, and stagnant phases—does not occur at random but, instead, is the reflection of underlying macroeconomic and financial forces driving the dynamics, i.e., growth patterns, interest rate cycles, inflationary pressure, and world geopolitical shifts (Hamilton, 1989; Alvi, 2018). Such "regimes" below will determine statistical character of data one is actually observing in finance. Therefore, the initial and simplest and most primitive step towards regime model formulation using well-specified HMM is to formulate well-specified macroeconomic model relating such hidden states to observable data in a way where the model will pick up structural information as distinct from noise statistics (Zucchini et al., 2016).

This part of the explanation prescribes the empirical research to be done and specifies four dimensions intertwined in the following way:

(1) The Role of Macroeconomic Analysis in Regime Detection:

Macroeconomic theory is the explanation that can tie loose ends of evidence together into a narrative of the business cycle. Data from past business cycles, documented by the National Bureau of Economic Research (NBER), enable us to make falsifiable hypotheses about which variables would respond most sensibly to regime change. Hamilton's (1989) first application of regime-switching models, for example, yielded compelling evidence that economic time series do have some breaks in behavior at some times in the business cycle. Theoretical underpinnings need to be employed in order to construct indicators not only in the market realm but also causal determinants of regime stability and regime change.

(2) Indicators Choice and Macroeconomic Financial Indicators:

Indicators choice is a serious effort at attempting to capture multi-dimensionality of the economy. An association of best variables should cut across different disciplines:

- Real Economic Activity: The economic activity is gauged by metrics that include the following: Gross Domestic Product (GDP) and manufacturing production.
- Health of Financial Markets: The S&P 500 (S&P 500 Index) broad equity indices are measured by the overall performance of businesses and investor sentiment.
- Inflationary Pressures: The Consumer Price Index (CPI) is tracked by taking a look at changes in the price level of a basket of consumer items.

- Monetary and Credit Conditions: The interest rates, for example, the US 10-Year Treasury Yield (US 10-Year Yield), signal the direction of monetary policy and market sentiment.
- Global Commodity Markets: The major commodity prices, like Crude Oil (Crude Oil WTI), are signals of world demand and geopolitical stability.

This multi-faceted strategy ensures the HMM an enlightened signal on which to base a prognosis of the underlying economic condition.

3. Dataset Retrieval and Sources:
The accuracy of any model is dependent on the quality of the input data. This involves acquiring consistent and dependable time-series data from credible sources such as the Federal Reserve Economic Data (FRED), the World Bank, and Bloomberg. Data frequency (monthly, quarterly), the history series length in order to obtain sufficient cycles, and revision treatment for previous data are some of the factors taken into consideration. A long, continuous dataset is needed to train the HMM so that it can recognize recurring regime patterns.

Raw economic data from a source is often unsuitable for direct modeling, as its accuracy can be seriously affected. It is important that a rigorous cleaning process be done to clean up the data, and this typically involves:

- **Handling Missing Data:** This involves a method where missing data can be handled by using appropriate imputation methods or backfilling so as to avoid introducing bias in the model.

These four steps form the foundation for accurate regime classification and forecasting for the model. When these are in place, a meticulously constructed dataset, guided by sound economic theory, will allow the HMM to effectively figure out the hidden states (Bear and Bull) that drive observable market phenomena.

The Role of Macroeconomic Research

Macroeconomic research provides the indispensable theoretical foundation for identifying and interpreting shifts in financial market regimes. These regimes are not abstract statistical constructs; they are manifestations of the underlying business cycle. For instance, economic expansions are typically associated with increasing GDP growth, rising equity markets, and lower credit spreads, reflecting optimism and easy financing conditions. Conversely, recessions are marked by falling asset prices, reduced consumption, and heightened volatility, signaling economic contraction and risk aversion. In modern financial modeling, purely statistical models are often insufficient; therefore, macroeconomic indicators are systematically combined with market-based signals to generate a more comprehensive and economically intuitive regime classification (Stock & Watson, 2016).

The HMMs better suit this task as their very form one-to-one maps this reality of the economy. The principle of work of HMM is the following way: an observational data sequence (say, returns from an asset, volumes traded)

is generated by an underlying unobservable sequence of discrete states, most commonly mapping one-to-one onto some economic cycles, say, expansion, contraction, or stagnation (Rabiner, 1989). The charm behind this piece is the embedding of the economic insight into the one probabilistic. Contended by Alvi (2018), speaking about the prediction of crude oil, so breathtaking was the explicit mapping from the observed dynamics out there in the market to the implicit regimes, where the latter had clear economic interpretations, from the detection of the patterns all the way to the inferral of the causality.

By analogy, in designing an effective regime-detecting process for equities/commodities, the macroeconomic indicators like inflation (CPI), interest rates like the US 10-year Treasury, and industrial productions offer the empirical anchor required to provide meaning to the otherwise invisible states uncovered by the model. One such invisible state with strong emissions from growth of the GDP and up surging equity indices confidently then perhaps could be imputed the meaning "bull" or "expansion" regime. Conversely, one state with high volatility, declining production, and expanding credit spreads perhaps is imputed the "bear" or "recession" regime. This inclusion guarantees that the derived model is not some statistical "black box" but an structured depiction of the economy where the hidden states have tangible meaning and the transition probability among them enforce the flow dynamics of the business cycle (Hamilton, 2016).

Indicator Identification

Selecting the appropriate indicators for a Hidden Markov Model (HMM) is a critical step that is capable of accurately differentiating between these regimes, i.e., bull, bear, and stagnant regimes. When choosing our variables, it is very important that they be sound theoretically, validated empirically, and capable of capturing the different phases of the different phase. These indicators will help us group them into two categories, namely macroeconomic indicators, which show the underlying health and state of the economy, and market-based indicators, which show how investors react sentimentally and real-time asset price dynamics.

1. Macroeconomic Indices

Macro indicators provide the background under which financial regimes take the course they do, informing us both on directionality as well as the health from the actual economy.

- **Gross Domestic Product (GDP):** Economy activity's alchemist stone, GDP growth rates are the benchmark for the regime classification. Even GDP growth is the bull regimes' signature, consisting of expanded corporate revenues along with robust consumer demand. Consecutive GDP contractions are the inevitable signature for the bear regimes, repeated official recessions (Hamilton, 1989). Static regimes are foretold by very sluggish growth rates around the zero point, consisting of an economy stalled by the transition process.
- **Inflation (Consumer Price Index - CPI, Producer Price Index - PPI):** The direction the rate inflation take are monetary policy determinants, by extension market regimes, by definition. Increase inflation, news particularly unexpected, depresses purchasing power and pushes the centers to demand the centers raise the rates the higher the potential trigger the onset the slope regime by causing the cost the capital

take off the economy latering. Lower rates, stable rates, however leads good bull market times, towards an dovish monetary policy. Stalemate regimes tend come with disinflationary periods, very low-inflation periods, the signal the economy'semand is slow the indecisiveness.

- **Interest Rates & The Yield Curve (Treasury Yields, Central Bank Base/Policing Rates):** They are good regime indicators. The slope of the level of the short rates to the level of the long rates has been convincingly proven to forecast recession (Estrella & Hardouvelis, 1991). The reversal curve yields (short rates over the long rates) is foreseen by an approaching regime of the bear, where the market is speculating economic weakness in the times ahead by the monetary authorities' downtrend. Bull regimes are frequent when the curves are upward-sloping, whereas flattening curve is associated with stagnant times.
- **Exchange Rate:** The unit of price of any nation, e.g., the US Dollar Index (USDX), are the indicators of external competitiveness as well as capital inflows to the universe total. The appreciating currency is the path to the "-flight to safety" out of the global bear markets, where the foreign capital is attracted by the currencies playing the role of the safe-haven. Or, the ever depreciating currency from the bull regime will have the tendency to take the ride over the revenues accruing from the revenues-based multinationals. Volatility is very pervasive under the regime of transition or stagnation out of the policy mess.

2. Market-Based Indicators

Whereas the macroeconomic indicators provide the underlying context, the market-based indicators provide the current, forward-looking measure of financial markets and investor sentiment. They are also useful for the forecasting of current dynamics of regime changes that have the ability to predict confirming macroeconomic data.

- Equity Index Returns (e.g., S&P 500, MSCI World): As the ultimate measure of the health of markets, price action within select equity indices represents the ultimate barometer of valuation shifts and investor sentiment. Repeated upward movements are the underlying feature of bull regimes, fueled by positivism and better earnings. Repeated losses are the definitive feature of bear regimes. Stable regimes are caused by the dramatic absence of directional momentum, with the indices fluctuating around an extremely narrow band, an indication of investor indecisiveness and balance between countervailing market forces.
- Volatility Indices (e.g., CBOE Volatility Index - VIX): The "fear gauge" is the term by which the VIX and other such indices have come to be referred. They measure the market's anticipated volatility over the near term. High and enduring volatility is one chief indicator for the bear regimes that indicate fear, uncertainty, andforced selling (Whaley, 2000). Observe also that inert regimes will have moderately high or surging volatility compared to the bull markets, as an indication of frustration and uncertainty but not necessarily the extreme degree present with full-blowncrises.

- Commodity Prices (i.e., Gold, Crude Oil): Cycles from commodities offer useful data surrounding supply and demand shocks globally. The price of crude oil is exceptionally informative as the price is also an indication of global industrial demand (a signal from the macroeconomic perspective) as well as supply disruptions from the impact of geopolitics (Alvi, 2018). Increases from the price of oil during an expansion are capable of triggering a bull market, whereas rapid slumps tend to predict declining global demand, predicting regime shifting to bear. Gold, an alternative currency itself, appreciates during periods of bear markets or extremely uncertain times, serving as a counter-cyclical signal.

By methodically merging the market-based with the macroeconomic indicators, from this aggregated procedure is born the HMM approach, as from the aggregated collection of observational signals the model will probabilistically deduce the sequence of the hidden regime states, thereby correlating apparent phenomena of market behavior to their economics origins.

Database Retrieval

Correct and systematic data extraction is the foundation for proper empirical financial and macroeconomic studies. The quality in any quantitative model, say the Hidden Markov Model (HMM), namely regime detection, is data accuracy input-based. Thus, the data are needed from official, consistent sources so that the raw dataset for the model is stable, reproducible, systematic bias-free. Sources utilized have an impact not only on data accuracy but also coverage throughout the period of time, frequency, as well as revision treatment, all considerations where longitudinal study becomes paramount (McCullough & Vinod, 1999).

A disciplined data source method is one of the utilization of several well-known data banks:

- **Federal Reserve Economic Data (FRED):** Developed by the Federal Reserve Bank of St. Louis, FRED is among the greatest resources on the U.S. macroeconomic data. It provides an incredible collection of time-series indicators including the Gross Domestic Product (GDP), the Consumer Price Index (CPI), the interest rates (federal funds rate, Treasury yields), as well as the employment data. Thanks to full historical coverage as well as the reliability, FRED remains an indispensable work among the working economists (Bureau et al., 2019).
- **Yahoo Finance, Bloomberg, Reuters:** Those are the industry benchmarks for historical as well as current financial market data. They offer the most basic data on stock indices (i.e., the S&P 500), single equities, as well as the indicators for volatility (i.e., the VIX). For programmatic access, the yfinance Python package also provides an easy-to-use API for the download of historical price data, for instance, the adjusted closing price and the traded volume.
- **International Monetary Fund (IMF) and World Bank:** The institutions provide systematically compiled cross-country macroeconomic indicators for the global-level studies. The datasets such as the World Economic Outlook by the IMF and the World Development Indicators by the World Bank are indispensable when carrying out studies on business cycles across the world, cross-country growth comparisons, as well as global capital flows.
- **Quandl (as Nasdaq Data Link today):** The platform pulls together an enormous collection of financial, economic, and alternate data from hundreds of sources. The platform also provides paid as well as free

datasets, so the platform is an extremely good one for data mining anything from the classic macroeconomic series to niche market data.

Data Cleaning Process

It is an unavoidable but sometimes convoluted process that transforms raw, often chaotic collections of data into standardized, orderly workable input to an HMM. The model's outcome is dependent entirely on the quality of this preprocessing, as the strongest algorithms will provide unreliable outcomes if furnished with faulty data. It involves a systematic order of procedures that have been dictated particularly to overcome the vices rampant among the macro-financial time series.

1. Handling Missing Values:

Economic and financial data are frequently tainted by discontinuities caused by publication delays, data constructions, or asynchronously traded weekdays. The favorite imputation method is particularly noteworthy, as poor methods have the potential to inject biasing/misspecify the time interdependencies. Some popular methods are:

- Forward Filling: As with the provided historical data, also straightforward but liable to spuriously smooth volatility.
- Linear Interpolation: Interpolation of unknowns from assumed straight line from the given data points, good for small gaps.
- Model-Based Imputation: Applying the more complex procedures, such as regression or expectation-maximization, to predict the imputed measures by their correlation with the balance from the quantitative variables under the prescribed, most stationarily formal method to all ((Little & Rubin, 2019).

2. Dealing with Outliers:

The financial time series also are also well-documented to have extreme values, also known as outliers, that are produced during the financial distress periods or illiquid markets. The outliers have too much influence on the HMM's parameter estimate so the model gets set off track by the tails. There are also some methodology's that are implemented to try to circumvent this:

- Winsorization: Truncation of extreme values at a predetermined percentile (e.g., 1st and 99th) of the data distribution.
- Z-Score Filtering: Detection and adjustment of observations that are more than a specified number of standard deviations away from the mean.
- Robust Transformations: Employment of statistics such as the median and interquartile range that are less sensitive to outliers than the mean and the standard deviation.

3. Frequency Alignment and Temporal Aggregation

Macro-financial data are multi-frequency in nature, i.e., GDP is published quarterly, inflation is published monthly, and equity prices are published daily. In building a consistent model, all series will need to be down-sampled to some common lower frequency (e.g., monthly). This involves:

- **Aggregation:** Transforming high-frequency data (e.g., daily stock returns) into lower frequency through end-of-period values, averages, or totals.
- **Interpolation:** The process of constructing higher-frequency estimates from lower-frequency data (i.e., constructing month-to-month estimates from quarter-to-quarter data), but something to be undertaken with caution since it can create spurious autocorrelation.

4. Stationarity and Comparability Transformations:

Most time-series models, including the HMMs, require or perform optimally with stationary data—series whose statistical properties (mean, variance) don't shift over time. And variables must be normalized in order to be comparable.

- **Log-Returns:** Price level non-stationarity is converted to log-returns, which have a greater chance of being stationary and can be directly interpreted as financial returns.
- **Differencing:** Trended macro series such as GDP or CPI are rendered stationary by calculating period-to-period differences (for instance, growth in GDP = GDP_t - GDP_{t-1}).
- **Scaling (Standardization):** Varying scale and unit variables (e.g., the GDP in billions and an interest rate in percentage points) are standardized by normalizing them to Z-scores. This ensures that no single variable dominates the inference of the model because it's large.

5. Test Stationarity:

Formal hypotheses of statistics are used to ensure that transformation process has actually attained stationarity. The Augmented Dickey-Fuller (ADF) test is most widely used, having a null hypothesis of the presence of a unit root (i.e., the series are not stationary). The KPSS test is its opposite, having a null hypothesis of stationarity. Both tests used as one combined verification is an extensive check, and a practice well entrenched in econometrics (Phillips & Perron, 1988).

6. Feature Engineering:

In addition to unprocessed numbers, engineered features can be constructed to provide the HMM with more informative signals. This improves the ability of the model to identify regimes by understanding dynamic market characteristics.

- **Moving Averages:** To suppress noise and uncover embedded trends.
- **Volatility Estimates:** E.g., rolling standard deviations of returns, to capture risk regimes explicitly.
- **Yield Curve Spreads:** Calculated as a differential of long- versus short-term rates, perhaps the most widely used measure of economic activity.

Combining Macro Indicators with HMM Structures

The ultimate aim of the thorough macroeconomic analysis and meticulous data preparation described in the previous pages is to prepare a dataset best adapted to the probabilistic structure of a Hidden Markov Model (HMM). This entails casting economic theory into quantitative, explicit form where statistical findings can be employed in some way to infer underlying dynamic states of the economy. This mix is illustrated very nicely by Alvi (2018) in the particular relationship between visible movements in oil prices and macroeconomic indicators like industrial production, using the HMM to uncover the underlying regimes underlying their co-movement. Here, under formalized arrangement, the preprocessed and cleaned indicators themselves now make up observable emissions, whereas unobservable, underlying economic states—"expansion," "recession," or "stagnation" forever remain the hidden states which the model will have to discover (Rabiner, 1989).

This letter provides an avenue for defining probabilistically segregated, regime-profiles uniquely segregated. For example, the regime-switching process might be defined by the following emission distributions:

- **Bull Regime (Expansion):** It is a state with high probability of generating signals such as robust positive GDP growth, positively stable return on equity, low or declining unemployment rate, and modest stable inflation. Joint occurrence of these signals suggests an economic growth and optimism phase.
- **Bear Regime (Recession/Contraction):** It implies high probabilities of adverse equity returns, rising unemployment level, high market volatility (as reflected by the VIX), and widening credit spreads as a sign of financial stress and risk aversion.
- **Stagnant Regime (Sideways/Transition):** This regime is required to provide low or zero GDP growth, low but steady equity returns, relatively high but stable unemployment, and overall lukewarm volatility, paralyzing an economy and a market in indecision or stabilization.

With regular incorporation of sanitized data sets from sources like pristine Bloomberg and FRED, researchers construct an integrative, multi-dimensional matrix of emissions for the HMM. Probabilistic inference can be empirically validated via the matrix. Then, the forward-backward algorithm can calculate the probability of the

observed sequence of data, the Viterbi algorithm can back out the most probable sequence of regimes over time, and the Baum-Welch algorithm can compute the parameters of the model—regime transition probabilities and emission probabilities of the indicators—sequentially from the data. All this effort, done in Step 1 of the project, translates raw economic data into a dynamic, probabilistic business cycle model.

Conclusion

much more challenging an assignment than is an exercise in data collection; it is the empirical pillars on which the entire regime detection model rests. This phase is the plain bridging necessary which interlinks, on the one hand, existing macroeconometric theory and, on the other, technical implementation of a Hidden Markov Model. It is theory-driven selective choice of indicators, extensive government database digging, and data scrubbing efforts performed with much effort for validity. It is this mixture of economic insight and computer brawn that distinguishes a statistically significant result from an economically significant result.

The importance of having such a firm foundation cannot be overstated. Without moral macroeconomic background and pre-preparation of data, the following HMM model development steps—training, validation, and regime detection—will be founded on sand. A well-trained inherited model based on poorly chosen, non-stationary, or inconsistent data might possibly be capable of discriminating between hidden states, but the states will be statistical abstractions, uninterpretable, and spurious in reality. For instance, an HMM would cluster the volatility episodes together but, in the absence of guidance from the coincident unemployment or GDP series, it will be impossible to distinguish a transient "flash crash" from an inherent "bear market" regime to an actual economic downturn.

Therefore, this first step is less of an introduction and more of a necessity. It ensures that the hidden states recognized by HMM are not random clusters but truly do have tangible, real economic meaning. By making the structure of the model contingent on business cycle dynamics, we can attribute meaning to a state by supposing that it is a "bull regime" in the sense of being statistically linked with high levels of GDP growth emissions and rising equity prices, or a "bear regime" in the sense of producing high volatility and spreading credit. This rigorous structure guarantees the model's output to not only be statistically correct, but also economically reasonable, with prudent conclusions to be drawn for finance forecasting and analysis.

The Regime Process and the Hidden Markov Model Implementation Introduction

The regime switching idea is a key idea of more recent econometric models of financial time series. The asset returns, the volatility process, and macroeconomic time series undergo regime switching between discrete regimes such as high-growth and low-growth regimes, bull and bear markets, or stormy and calm volatility regimes. Hidden Markov Models (HMMs) provide a simple statistical framework for describing the regime dynamics. By taking the system to go through a sequence of hidden states every defining the distribution over the observables, HMMs allow researchers to represent complex, non-linear processes in a tractable way (Rabiner, 1989; Zucchini et al., 2016).

This section formally and empirically defines the regime process. It does so, specifically, by considering how regime switching is measured in the HMM model, mathematical formulation of the regime process, and how the proper code in the hmms library is understood in use. The exposition connects formal HMM formulation with economic interpretation of regime switching and therefore unites theoretical abstractions and empirical application.

Theoretical Background of Regime Processes

Regime process is the name given to a transition driving force of transitions from latent states in a time series model. The regime process for HMMs is a discrete-time, finite-state Markov chain. The following are the typical properties of this Markov chain:

1. **Finite State Space:** Let $S = \{1, 2, \ldots, K\}$ be the set of regimes. There is one regime for every possible state of the economy or the market, i.e., a high-volatility regime and a low-volatility regime.
2. **Markov Property:** Markov Property: The probability of moving to a new regime depends only on the current regime and not on the entire history of past regimes. Algebraically:

$$P(s_t = j \mid s_{t-1} = i, s_{t-2}, \ldots, s_0) = P(s_t = j \mid s_{t-1} = i) = a_{ij},$$

Where a_{ij} denotes the transition probability from state i to state j.

3. **Transition Probability Matrix (TPM):** The regime process is fully characterized by a transition probability matrix, whose elements satisfy: $K \times K$

$$a_{ij} = P(s_t = j \mid s_{t-1} = i), \quad \sum_{j=1}^{K} a_{ij} = 1.$$

4. **Stationary Distribution:** The stationary distribution of the Markov chain does exist, if the chain is ergodic (irreducible and a periodic and satisfies. The stationary distribution π, satisfying $\pi A = \pi$. is the long-run probability to be in some regime.

This Markov process underlines HMMs' regime process in linking the observable time series data and hidden states using emission probabilities. For example, in finance modeling, a "high-volatility" state may generate returns from a high-variance distribution whereas a "low-volatility" state is synonymous with one of low variance (Hamilton, 1989).

Mathematical Representation of the Regime Process in HMMs

The regime process can be represented mathematically as:

- Let $s_t \in S$ denote the hidden state (or regime) at time ttt.
- The sequence of hidden states $\{s_t\}_{t=1}^{T}$ evolves according to a first-order Markov chain with transition matrix A.
- The initial state distribution is represented by $\pi_0 = [\pi_1, \pi_2, \ldots, \pi_K], \quad \text{where} \quad \pi_i = P(s_1 = i).$

Therefore, the probability of observing a particular regime sequence is:

$$P(s_1, s_2, \ldots, s_T) = \pi_{s_1} \prod_{t=2}^{T} a_{s_{t-1}, s_t}.$$

This specification points out the stochastic and path-dependent nature of the process for regimes because of the randomness of the sequence of the hidden states.

Practically speaking, the estimation of the transition probability is key to the understanding of regime process dynamics. Suppose one obtains an estimated probability a_{12} (transition from regime 1 to regime 2) of large value. Then the system will switch very often between regimes. Large diagonal elements a_{ii} signify persistence in the sense that the system, having entered a regime, will tend to remain longer in the regime.

Economic Theory of Regime Switching

- Regime processes also have a natural economic theory within financial econometrics. For instance:
- Bull and Bear Markets: Asset returns may possess varying statistical properties, depending on whether the regime is bullish (increasing) or bearish (declining) (Guidolin & Timmermann, 2005).
- Business Cycles: Macroeconomic variables such as expansion in GDP can have varying dynamics during boom and bust (Hamilton, 1989).
- Volatility Clustering: Time series of stock returns show episodes of high and low volatility, consistent with a regime-**switching variance process (Ang & Bekaert, 2002).**

By treating these regimes as an explicit model component, HMMs enable estimation of the probability to be in either of the regimes, make the probability of future regime shifting estimable, and incorporate such dynamics into forecasting and decision-making.

Use of Regime Processes from the hmms Library

The Python package hmms gives an open and robust Hidden Markov Model framework. States can be defined, transition matrices defined, and model parameters updated from observations using this package. The regime process is being executed stepwise in this section.

1. Definition of State and Transition Matrix

The first action is to define the number of hidden states (regimes) and the structure of the transition matrix. For hmms, one might do this in the following way:

2. Estimating Transition Probabilities

In practice, transition probabilities are estimated by maximum likelihood methods, specifically the Expectation-Maximization (EM) algorithm (Baum-Welch algorithm). Estimation is done automatically by the hmms package:

This step distills the transition matrix from the data observed, obtaining empirical regime dynamics.

3. Decoding the Regime Sequence

Once the model has been estimated, the Viterbi algorithm can be used to discover the most likely regime sequence. This provides a detection of when regime switches occurred in the data:

4. Regime Probabilities

Alternatively, the model can forecast posterior probabilities of being in each regime at each time step. This is used for probabilistic instead of binary classification of regimes:

Challenges of Modeling Regime Processes

Although Hidden Markov Models (HMMs) are a powerful and expressive model for sequential data with latent structure, their practical usage is plagued by several serious problems that have to be adequately addressed by researchers in order to give valid and meaningful results.

1. Model Selection and Number of Regimes

Part of the easy but not-so-obvious problem is determining the number of hidden states, K. It's a standard trade-off in statistical modeling. Selecting too few states (underfitting) results in a too-simple model that is unable to capture the entire richness and complexity of the underlying economic dynamics, perhaps confounding two different regimes such as a "stagnant" phase with a mild "recession." Selecting too many states (overfitting) results in a too-perfect model that fits sample data too well, capturing spurious, non-repeated regimes that do not generalize well to new data and are non-parsimonious (Zucchini et al., 2016). Although information criteria like Akaike Information Criterion (AIC) or Bayesian Information Criterion (BIC) generally are employed to make this decision, ultimate selection would have to be guided by economic theory in addition to the state interpretability.

2. Parameter Sensitivity and Estimation

The Baum-Welch algorithm works on the basis of the Expectation-Maximization (EM) algorithm, which is convergent but to local maximum of the likelihood function. Parameters ultimately obtained—e.g., transition matrix and emission distributions—may be highly dependent on initializations. With poor initializations, the algorithm can converge to a suboptimal low-quality solution which is not an admissible model to the underlying data-generating process. This involves a method of repeated random re-starts or re-start with heuristic initialization in order to reach the best possibility of finding the global, or at least improved, optimum.

3. Interpretability of Latent States:

One benefit of using HMMs in econometrics is that they possess the clear meaning of being economically interpretation endowed. If states are arrived at statistically strictly rather than theoretically, however, their meaning is in jeopardy. Such a period of negative return high volatility can be identified by a "financial crisis" or a "sharp recession," and the two are to be distinguished by looking at the hard emissions (i.e., the credit spread

or industrial production record over time). The modeler would then need to label and categorize the states actively after estimation by their relative emission parameters and translate them into economic intuition.

4. Non-Stationarity and Time-Varying Dynamics:

Time-homogeneity (time-invariance) is enforced on the state transition probability in standard HMM. This is not typically met for financial and economic time series for which long-term determinants such as level of debt increasing or regime shift at the central bank may render recession entry probabilities a function of. To this aim, new additions have been proposed, including the Time-Varying Transition Probability (TVTP) HMM, where the transition matrix can be a function of observed covariates (Diebold et al., 1994). This expansion-recession transition probability can, for example, be made a function of the slope of the yield curve.

Conclusion

Regime process in Hidden Markov Models is a mathematically founded method for economic and financial time series structural change modeling. Regime switching stochastic persistence modeling between regimes such as bull/bear markets or high/low volatility regimes, HMMs model the system using a hidden Markov chain. Python hmms library is an easy-to-use package for economic interpretation, regime sequence decoding, and model estimation. Despite being marred by model selection and parameter estimation problems, the regime process paves the way in the art of successful empirical use of HMMs.

pgmpy Training of Bayesian Network and Parameter Test/Validation Introduction

Probabilistic Graphical Models (PGMs) offer a principled representation of intricate distributions by taking advantage of graph structure, in which nodes are random variables and edges are dependency relations between them (Koller and Friedman 2009). Hidden Markov Models (HMMs), which have previously been mentioned in previous phases of this project, are a subclass of Dynamic Bayesian Networks (DBNs). In addition, Bayesian Networks (BNs) provide support for modeling temporal and cross-sectional interdependence between the variables, providing a stronger foundation for regime change modeling for macroeconomic and financial time series (Murphy 2012).

The pgmpy Python package provides strong support for training and constructing Bayesian Networks. This part deals with training such networks, what parameter estimation techniques one can use, and how validation and testing strategies are employed to make the model strong. The part is built from Alvi (2018) straight but using contemporary computing resources and research.

Bayesian Networks in Financial Regime Detection

Bayesian Networks are directed acyclic graphs (DAGs) to represent conditional dependencies between random variables. For regime classification, a BN may be used for estimating the probabilistic linkage of macroeconomic indicators (e.g., GDP, unemployment, inflation) to regimes in the market (bull, bear, stagnant). This also alleviates researchers from sequential constraints of HMM and considers exogenous, contemporaneous variables in regime modelling (Friedman et al. 1999).

For example, to predict the prices of crude oil, a Bayesian Network can explain regime dependence of the oil price on previous regimes, as well as on global demand conditions, supply shocks, and exchange rates. This makes the BN more general compared to the traditional Markovian models, where the next period's state is a function of just the current hidden state (Hamilton 1989).

Structured Learning

Bayesian Network structure learning is the first phase of Bayesian Network construction in which DAG of variable dependencies is learnt. Three ways of doing so are:

1. **Constraint-based learning:** Conditional independencies between the variables are queried by deriving algorithms such as the PC algorithm in order to reason about dependencies (Spirtes, Glymour, and Scheines 2000).
2. **Score-based learning**: Candidate networks are scored here on the basis of metrics such as Bayesian Information Criterion (BIC) or Akaike Information Criterion (AIC), and then the best-scoring network is selected (Heckerman 1995).
3. **Hybrid methods:** These combine constraint-based and score-based methods for more effective structure learning.

In practice, with financial time series, score-based approaches are superior for noisy and high-dimensional data (Koller and Friedman 2009). With crude oil prices, stock market indices, and inflation statistics, for example, a score-based approach might prefer networks that optimize the trade-off between explanation and model complexity.

Parameter Learning

Once the structure is established, parameter learning learns the nodes' conditional probability distributions (CPDs).

1. **Maximum Likelihood Estimation (MLE):** Recovers parameters by directly learning frequencies from data. There tends to be a Gaussian assumption in continuous variables (Murphy 2012).
2. **Bayesian Estimation:** Establishes prior distributions over the parameters and updates with data to make the estimates strong even in sparse data (Heckerman 1995).
3. **Expectation-Maximization (EM):** It works best if the data contain missing variables or unobserved variables. EM offers iterative missing value and parameter estimates until it converges (Dempster, Laird, and Rubin 1977).

In finance applications, EM is useful because regime states are not all observed. We, for example, never observe the true "bull" or "bear" states but estimate them from volatility and market returns.

Testing and Validation

Validation to guarantee that the Bayesian Network is able to perform well on more than the training sample. Testing is in several different forms:

1. **Log-likelihood and Perplexity:** These measures test good fit of the model over new data. Good fit is characterized by high likely value (Murphy 2012).
2. **Cross-validation:** Train-test set partition avoids overfitting noise from the model.

3. **Information Criteria (BIC/AIC):** Applied in structure learning to achieve a trade-off between fit and model complexity (Koller and Friedman 2009).
4. **Posterior Predictive Checks:** Model sample from the sample and compare against observed distributions (Gelman et al. 2013).

Validation is more and more necessary in the classification of regimes because regime misclassification will yield erroneous investment decisions. For instance, misclassification of the bear regime will result in defensive action that will fail to match market return.

Challenges in Bayesian Network Training

While Hidden Markov Models (HMMs) give a robust and universal description of sequential data whose structures are unexplored, empirical applications of such models are accompanied by a set of unwished-for problems that researchers have to solve in minute details in an effort to attain credible and interpretable outcomes.

1. Model Choice and Number of Regimes

One of the simplest and least intuitive ones is to determine the number of hidden states, K. It is a typical trade-off in statistical modeling. Selecting too few states (underfitting) results in an overly simple model that does not capture all the richness and texture of the underlying economic forces, perhaps merging regimes with distinct attributes like a "stagnant" regime and a mild "recession." Or, selecting too many states (overfitting) results in a model that over-fits the sample observations, capturing non-recurring, spurious regimes that do not generalize well to future observations and are non-parsimonious (Zucchini et al., 2016). Although the choice criteria like the prediction criteria of Akaike Information Criterion (AIC) or Bayesian Information Criterion (BIC) are systematically applied in this decision, the decision must also be made considering economic theory and interpretability of the obtained states.

2. Parameter Identification and Estimation Sensitivity:

The Expectation-Maximization algorithm applied in the Baum-Welch algorithm converges to a local maximum of the likelihood function guaranteed. Therefore, the last estimated parameters—e.g., the transition matrix and the emission distributions—will be very sensitive to initializations. Poor initialization would lead the algorithm to converge to a certain suboptimal solution that is a poor estimate of the data-generating process. This necessitates a culture of periodic re-starting several times or the supplying of expert, heuristic-generated starting values hoping to maximize the chances of converging to the global, or at least better, optimum.

3. Interpretability of Latent States:

One of the econometrics' strongest points for HMMs is economic interpretation in a clear-cut manner. However, when states are derived solely through statistical optimisation and not via derivation theory, it is not clear immediately how one should interpret them. A high-volatility, negative-return state could quite possibly be a "financial crisis" or a "sharp recession" and can be differentiated only by examination of the respective emissions (e.g., credit spread behavior or industrial production behavior). The modeler would then characterize and name states post-estimation by examination of their resulting emission parameters in a manner that connects them to conventional economic concepts.

4. Non-Stationarity and Time-Varying Dynamics:

The simplest HMM assumes no change in transition probabilities between states over time (time-homogeneity). This would be more likely than not to be a false assumption for economic and financial time series, where the likelihood of entering recession may be subject to long-run influences like rising indebtedness or shifting central bank regimes. To avoid this, more recent extensions have been suggested, e.g., the Time-Varying Transition Probability (TVTP) HMM where the transition matrix is allowed to vary as a function of observed covariates (Diebold et al., 1994). For example, expansion-to-recession transition can be made a function of the slope of the yield curve.

Case Study: Bayesian Network based Regime Detection

For instance, forecasting oil price regimes from macro indicators. When BN contains nodes like OilPrice, ExchangeRate, Inflation, and MarketRegime. It can be learned by structure learning and parameter estimation that there is a bear regime of hard dollar and high oil price. It can aid hedging strategy of energy companies and investors (Alvi 2018).

Empirical research, for instance, by Song et al. (2016), has shown Bayesian methods to be superior to the conventional econometric method to detect regime changes. The method of Alvi can be generalized, however, to a more multivariate context through the inclusion of pgmpy's learning capability.

Conclusion

Bayesian Networks such as pgmpy are a superb class of Hidden Markov Models for identifying financial regimes. With learning structure, learning parameters, and scoring, such models are a platform where one can model cross-sectional and macro and market data time-series dependence. Beaten black and blue with beset by

overfitting and non-stationarity, BNs are a goldmine in probabilistic forecasting and decision-making in finance. With computer resources and statistical programs at one's fingertips, it is not as challenging for researchers to utilize whatever was possible in Alvi (2018) to come up with models that have not just theoretical significance but empirical significance as well.

Step 5

Group Narrative (Integrated Papers)

Introduction

This essay benefits from the collective efforts of three work on complementary components of financial regime detection using the two methodological approaches of Hidden Markov Models (HMMs) and Bayesian Networks (BNs). The project is structured around three distinct yet related areas of research, each providing a missing piece to the overall analytic approach. The contributions are: macroeconomic research background research, indicator selection, and data preprocessing; regime process analysis thereafter and computational implementation of HMMs and training, building, and validation of Bayesian Networks using the pgmpy library . Separately from one another, these are works tackling discrete issues; together, they constitute an integrating pipeline between methodologically sound implementation and practice and abstract economic theory and expected financial regime in a straightforward manner. This integration is the direct result of the regime-switching groundwork laid by economists like Alvi (2018), and continues to advance the frontier even further with the use of advanced computational tools and state-of-the-art statistical investigation.

The volume of individual intellectual work that has been expended on this co-authored project is the equivalent of about 6,000 words of writing to define methodology and literature review sections of an M.A.-level research report or mini-capstone. The organization is intentional, moving outward from empirical source to general theory and technical specificity. The exercise demonstrates the interdisciplinarity of contemporary financial econometrics to depict regime modeling an impossible task to contain within any one discipline. It needs economic theory application to define the problem space and select variables, statistical modeling for enabling regime volatility and interaction, and machine learning deployment in order to perform correct estimation of intricate models and test them on data. A three-fold approach in itself guarantees that the resulting models not only statistically perform but are economically sound and computationally manageable.

Section 1: Macroeconomic Analysis, Determination of Indicators, and Preparation of Data

Identification of the financial regime must be more than a statistical exercise in isolation of its macroeconomic context. The bull, bear, and stagnant unobserved states are not random; they are emergent properties of economic fundamentals. Key macroeconomic forces such as Gross Domestic Product (GDP) growth, inflation pressure, unemployment rate, and monetary policy are the motive forces of structural conditions offering

support or weakness to medium and long-horizon financial asset prices (Hamilton, 1989). They are the "why" of market regimes, forecasting economic booms, downturns, or stagnations.

In contrast, financial market information convey high-frequency lead indicator messages of anticipated real economy adjustment. Large equity index return benchmarks (e.g., S&P 500), implied volatility benchmarks (e.g., the VIX), credit spreads, and relevant commodity prices convey aggregate sentiment and opinion of market participants today. Macroeconomic indicators write the story below, but financial data capture the here-and-now ledger of market euphoria or fear.

The methodological brilliance of this section is in the sensitivity to include both sorts of data. A model based on macroeconomic data only will lag to signal a regime change due to publication lags, while a model based on financial data only will be prone to spurious regimes driven by sentiment rather than fundamentals. By putting them together, we create a better and more multi-faceted set of observable emissions for the HMM. The model is then able to distinguish, say, a short-term financial adjustment from a genuine bear regime because of a slowing economy. This hybrid methodology, one which aligns observable signals with hidden economic regimes, is essential in identifying and interpreting regime shifts among bull, bear, and stagnation regimes (Stock & Watson, 2016).

Significance of Macroeconomic Indicators

Financial indicators are closest related exogenous variables slicing deepest into financial market action. They provide quantifiability of underlying state of economy indirectly influencing profitability of corporate sub-segments, investor sentiment, and monetary policy. For instance, as rising and consistent GDP growth and falling unemployment, the profitability of companies rises, the confidence of the investors rises, and the kind of setting that will bring regime shift to bull market (Fama & French, 1989). But a rise in inflation reduces real returns and pushes central banks out of money policy through rising interest rates. This will have the effect of raising cost of capital for consumers and businesses and also risking deflationary patterns of economic that are bound to drive financial markets into a bear trend cycle of declining prices and risk aversion.

The economic systems today are internationalized and interdependent, and their employment of local data in regime identification consequently is increasingly problematic. Any credible regime identification model simply is not possible to build on local data due to synchronized business cycles, cross-border trade, cross-border capital flows, which cause economic trends and shocks to spill over very rapidly over the border. Not only must statistical offices of countries release data, but so must authoritative global world institutions. These institutions like the International Monetary Fund (IMF), the World Bank, and the Organisation for Economic Co-operation and Development (OECD) release homogeneous, homogeneous, and comparable world-wide macroeconomic performance series. Global data along with market-level high-frequency data can be used to construct complete models which one can, in turn, use for the estimation of domestic as well as global spillover terms to construct an ever-richer and richer regime detection model.

Indicator Choice

The indicators have been selected due to their theoretical and empirical associations with business cycle and financial market performance. The selected variables form a well-delineated set which equips the HMM with a sufficient number of hints to enable it to recognize several economic regimes. The most critical indicators used in the model are:

- **GDP Growth:** The most and best measure of the economy of a nation, rising Gross Domestic Product is the most transparent sign of busts and booms. Positive growth consistently takes place under bull regimes, and two successive quarters of negative growth technically qualify as a recession and are typical of bear regimes (Hamilton, 1989).
- **Unemployment Rate:** A lagging but highly informative indicator of the overall health of the labor market. Declining unemployment sows seeds of consumer confidence and purchasing power that have been translated into economic booms. Unemployment rises sharply, then it is a sure bet that a bear regime of contraction, economic ill health, and shrinking aggregate demand is at hand.
- **Inflation (PPI, CPI):** Consumer Price Index (CPI) and Producer Price Index (PPI) record a trend in the price level. Increasing and high inflation erodes disposable income and purchasing power and makes central banks adopt tight monetary policy and thus create a bear market. Low and stable inflation works best to support long bull markets.
- **Interest Rates (Treasury Yields, Fed Funds Rate):** Interest rates influence asset prices directly in discount rate procedures and investor appetite for risk premium. Federal Funds Rate specifies short-term cost of capital, and long-term Treasury Yields track market growth and inflation expectation. A specific type of yield curve inverts as a good indicator of recession (Estrella & Mishkin, 1998).
- **Prices of Crude Oil:** Being the world's strongest commodity, crude oil price is a first-order predictor of supply and demand shock in the world. Gross oil prices have direct effects on the cost of production, corporate profitability levels, and even on world trade balances. For this reason, oil price movements are a significant input into detecting regime changes in regimes of the world economy (Alvi, 2018).
- **Equity Market Indices (like S&P 500):** The stock indices reflect the optimism and expectations of investors about future economic activity and corporate profits. Lovely uptrends would more probably be indicating bull regimes, and bear regimes would have recurring downtrends. Their presence is extremely frequent, relaying instant indications of changing market conditions.

Dataset Retrieval

The success of any finance empirical model hinges on the credibility, reliability, and quality of its raw data. For modern computational finance, this implies a data-retrieval system firmly rooted in Application Programming

Interfaces (APIs) and meticulously curated open databases. Data consistency, integrity, and transparency are assured in the system. Primary sources constitute the core of this data collection system:

- **Yahoo Finance (via the yfinance Python library):** A publicly available data source of high-frequency financial market data, i.e., daily historical equity prices (open, high, low, close, volume) of large indexes like the S&P 500 and large commodity futures like Crude Oil WTI.
- **FRED (Federal Reserve Economic Data):** Sponsored by the Federal Reserve Bank of St. Louis, FRED is the authority on U.S. macroeconomic data. It provides official, normalized time series for all the major indicators like Gross Domestic Product (GDP), the Consumer Price Index (CPI), unemployment rates, and Treasury yields, and is a must-have for macroeconomic modeling (Bureau et al., 2019).
- **IMF Data & World Bank Open Data:** Both organizations provide broad, standardized datasets for application in modeling global macroeconomics. The International Monetary Fund (IMF) provides sovereign debt data and international financial statistics and the World Bank provides cross-country GDP data, poverty data, and development indicators, the building blocks with which to construct models of global economic relationships.
- **Quandl (now Nasdaq Data Link):** This is a vendor that provides a broad financial, economic, and alternative data, both public and commercial sources, so one-stop for everything from standard fundamentals to more unusual data.

Automating data retrieval is accomplished through the use of Python libraries such as pandas_datareader (which offers an entry point to a single interface to FRED and others), yfinance, and the general requests library to issue actual API calls directly. This kind of scripting of data gathering is very much worth performing both for reproducibility, in the sense that the entire data pipeline can be automated and run again and again, and scalability, in the sense that modifications to datasets and additional variables can be safely achieved without being dependent on human intervention.

Data Transformation and Cleaning

Raw financial and macroeconomic data are never, if ever, directly usable by quantitative models since they contain missing values, outliers, and other forms of noise. A preprocessing pipeline that is robust should therefore be put in place to be able to take such raw data and preprocess it into a clean form that is analytically usable. This eliminates both the potential for biased parameter estimates and the issue of the model responding to artifacts in the data rather than economic variables. Some of the most critical steps are involved in preprocessing:

1. **Handling of missing values:** Economic time series tend to be incomplete due to delayed publication or revision. Deletion of these values would lead to serious bias. Instead, imputation methods are employed, e.g., forward-filling for minor gaps or more sophisticated interpolation methods (e.g., linear, spline) that extrapolate missing values from the trend in the supporting data series.

2. **Identification and Treatment of Outliers:** Financial time series record huge fluctuations, especially in times of crises. The outliers disproportionately affect the model parameters to a great degree. They are detected by using robust statistical methods, i.e., Z-score trimming or Interquartile Range (IQR) procedures. Outliers are treated by winsorization (removal of extremes) or transformation to eliminate their undesirable impact.
3. **Transformation to Stationarity:** Financial price series are drifts and volatility-changing non-stationary. The simplest transformation is to express prices as log returns by taking their log such that prices are in log returns. The operation, difference in natural log of prices across periods, makes variance not time-varying and leads to a series near stationarity—a typical time series model assumption (Tsay, 2010).
4. **Scaling and Normalization:** The variables of the data in a multi-indicator model (for example, GDP in billions, percentage interest rate, and volatility index) are very differently scaled. To prevent the larger-ranging variables from overpowering the model's inference, Standardization (i.e., to Z-scores) or Min-Max Scaling is used. By doing this, all of the features have the same potential to contribute to the learning algorithm.

Together, the end-to-end pre-processing sequence ensures that inputs to representatives of machine learning and models of the underlying financial and economic dynamics, however unbiased they prove to be, are precise and therefore constitute a sound foundation for effective regime detection and prediction (Tsay, 2010).

Section 2: Regime Processes and Hidden Markov Models

Hidden Markov Models (HMMs) offer a mathematically advanced, probabilistic explanation of how one determines hidden or latent regimes underlying allegedly observed patterns in financial markets. The hypothesis is that highly complicated financial time series—stock returns, volatility, and volume, for example—are not produced by a stationary process. Rather, they are manifestations of some finite number of discrete dynamic regimes—bull (expansion), bear (contraction), and stagnant (sideways/transition) markets.

These said regimes are the implicit regimes of the model. No direct information is provided to the model regarding whether on a given day the market is in a bull or a bear regime; it must derive it from observable data. These are hidden states, i.e., state-dependent in formation, i.e., probability of arriving at any future state is a function of current state but not the entire history (Rabiner, 1989). This is expressed in the form of the state transition matrix.

There is one additional independent free-standing emission probability distribution across latent states. As an illustration, a bull state may be a positive mean but small variance asset return probability distribution, and a bear state may be a negative mean but greater variance. HMM algorithm, employing techniques such as Forward-Backward and Viterbi algorithms, uses the observable sequence of observations to compute three central problems: observation probability scoring computation, most likely sequence of underlying states

estimation, and learning observation-dependent model parameters (transition and emission probability) from observations. HMMs are positioned with the best estimate of unobservable economic fundamentals and sentiment that come in the form of observable regime shifts in financial data.

Theoretical Background

An HMM consists of:

- A set of hidden states $S = \{s_1, s_2, \ldots, s_N\}$.
- Transition probabilities between states $a_{ij} = P(s_t = j \mid s_{t-1} = i)$.

Emission probabilities that describe the generation of observed data from hidden states.
In finance, the observed data are typically returns or volatility measures, and the hidden states are market regimes.

Algorithms

The power of the Hidden Markov Model (HMM) framework is realized by a collection of special-purpose algorithms, each of which solves a fundamental problem in the application and training of the model. The algorithms function in concert to enable inference, decoding, and learning in sequential data.

- **Forward Algorithm:** The algorithm of dynamic programming is computing the solution to the evaluation problem. It is computing the probability of an observation sequence given the hypothesized economically model HMM parameters. Propagating the probabilities recursively downstream in the sequence, it estimates the probability of being in some particular state at time $$t$$, based on all the observations through time $$t$$, without the computational practicability of enumerating all possible state sequences (Rabiner, 1989). This is used in model scoring and comparison.
- **Backward Algorithm:** In the backward direction, the Backward Algorithm is seldom alone but a fundamental component of other computations. It computes the probability of the future sequence of observations for a particular state at time *t*. When combined with the result of the Forward Algorithm, it enables calculation of a probability of being in some specific state at some specific time given the entire observation sequence, a fundamental component of learning (Murphy, 2012).
- **Viterbi Algorithm:** This dynamic programming algorithm solves the problem of decoding—how to find the single most likely sequence of hidden states that could have generated the observed data. Unlike the Forward Algorithm, which calculates the sum of the probabilities, the Viterbi Algorithm traces the maximum probability path at each step, choosing the best path through the lattice of possible states (Forney, 1973). This is the initial phase of historical regime classification, a sharp "best guess" of the condition of the market for each day.

- **Baum-Welch Algorithm:** Baum-Welch algorithm is a particular case of the Expectation-Maximization (EM) algorithm and uses the Baum-Welch algorithm to find the solution of the learning problem. Baum-Welch algorithm is used in the estimation of the state transition matrix and the emission probability distributions of the HMM from an unlabeled observation sequence. It cycles through an Expectation (E) step, in which it computes expected counts of states as a function of the existing parameters (using the Forward-Backward algorithm), and a Maximization (M) step, in which it maximizes the parameters for the purpose of making the provided expected counts more probable (Rabiner, 1989). This makes it possible for the model to learn the behavior of the different regimes as a function of the data directly.

Regime Typology

Hidden Markov Models are particularly suited to the identification of financial regimes since they possess an inbuilt ability to detect two of the most significant features of financial markets: state persistence and probabilistic shifting of states of market conditions. Financial markets do not switch state randomly but follow periods of persistence where a single setting or environment will dominate, and then they will switch abruptly or stepwise to another state. This dynamics is explicitly encoded by the topology of the HMM, i.e., its state transition matrix and state-dependent emission distributions.

The power of the model lies in the capability of statistically detecting and distinguishing among such latent states based on observable observations. One of the regimes anticipated can be described as follows in the HMM model:

- **Bull Regimes:** These are steady economic boom and euphoria regimes. Quantitatively, the emission distribution of a bull regime is one with positive mean for asset returns, normally with smaller variance, i.e., smooth ascending trend with relatively stable price behavior (Guidolin & Timmermann, 2007). The model considers that those observations which clustered around these values have high probability to be emitted by the "bull" latent state.
- **Bear Regimes:** On the other hand, bear regimes are periods of worsening economic happiness and pessimism. The HMM classifies such periods as a negative mean distribution of emissions and significantly higher variance. This describes early bear market trends: consistent price erosion and increasing uncertainty and volatility (Ang & Timmermann, 2012).
- **Stagnant Regimes:** They are a state of transition or indecisiveness with the market and where there is neither trend of increase nor decline. The emissions pattern corresponding to it would then be a one with nominally zero mean return but maybe with moderate variance, and this is the best manner in which to describe a directionless choppy price movement with zero possibility of bear market or anticipation of bull market.

With encoding of intertransition probabilities between single emission parameters and themselves, HMM offers dynamic and probabilistic characterization of market evolution with a wholesome advantage over static model in capturing time-varying nature of financial data-generating processes.

Strengths and Weaknesses

Hidden Markov Model (HMM) in the application in the detection of financial regimes has categorical advantages but is faced with insurmountable weaknesses with which one must acquaint himself.

Strengths:

1. **Catches State Persistence and State Transitions:** Somewhat surprisingly, perhaps the most significant benefit of HMMs is that they catch state persistence and state transitions. The state transition matrix will then be able to distinguish financial markets' propensity to remain in one regime (e.g., a bull market) for some amount of time before transitioning into another one. This allows the model to address the well-documented phenomenon of regime "stickiness" and forecast the likelihood of an upcoming transition that can be useful to asset allocation strategy and risk management.
2. **Provides a Probabilistic Inference Model:** Unlike hard-clustering models with one state assignment per time step, HMMs are provided in a complete probabilistic model. They provide a probability distribution of the states of each time step, providing some uncertainty on the regime label. It is extremely useful when financial regimes are unclear and this probabilistic output is directly used to inform higher-level decision models.

Limitations

1. **Sensitivity of Initialization and Model Specification:** Local convergence of Expectation-Maximization algorithm used to train (Baum-Welch) to a maximum learned model parameters can thus be highly sensitive to initial values assumed by the transition and emission matrices. Also, it's not simple to select the size of the hidden states (K); the incorrect selection will lead to overfitting (too many states) or a very unsophisticated model that won't be able to pick up significant dynamics (too few states).
2. **Structural Simplifications:** Most elementary of HMMs calls upon a first-order Markov process to state building when the succeeding state depends only on the current one. That is too simplistic to express real financial activity, where state duration or dependence on order higher than one has some effect on transition probability (Cappé, Moulines, & Rydén, 2005). Furthermore, the base HMM has time-homogeneous assumptions regarding the transition probabilities independent of time, usually quite far from reality if financial conditions are time-dependent. HSMMs and Time-Varying HMMs have been suggested to circumvent such specific restrictions.

Section 3: Bayesian Networks, Training with pgmpy, and Validation

While Hidden Markov Models (HMMs) offer a thrilling model of regime detection across time, they themselves are constrained by the Markovian assumption to limiting more richer dependency structures for variables to limiting to temporal dimensions. Bayesian Networks (BNs) are more general and more versatile as a sub-class of Probabilistic Graphical Models (PGMs) and less constrained in avoiding doing so by being capable of expressing much richer structures of variable dependencies (Koller & Friedman, 2009).

The most noteworthy difference is topology of the relations that they can be used to represent. A subfamily HMM is a time-structured BN of one specific type with a chain-topology hidden state and the observation that the observations are conditionally independent within a state. A general BN does not have so many constraints. It also has the ability to store an interweave of relationships between history and time (e.g., yesterday's volatility in today's market regime vs. today's surprise inflation announcement effect on market regime, bond yields, and the exchange rate simultaneously).

This flexibility is incredibly strong in explicitly incorporating exogenous macroeconomic variables in the regime detection model. For example, a BN, e.g., can explicitly specify causal or influence channels, i.e., how a central bank interest rate decision (an exogenous macroeconomic variable) must affect the market regime, which affects equity returns and credit spreads (Murphy, 2012). It enables the model to ask advanced conditional questions, e.g., predict the likelihood of a bear regime if inflation has increased by 2% and the yield curve has inverted. Modeling these difficult, non-Markovian dependencies, BNs enable a more expressive, causality-aware description of the determinants of financial regimes.

Structured Learning

Structure

There structure learning, which is one of the main issues of creating a Bayesian Network (BN), is to learn an optimum-directed acyclic graph (DAG) of probabilistic dependencies among variables from a data set. It does not seem a very simple task as there are super-exponentially many possible DAGs when increasing variables. Main directions to the solution are:

Constraint-Based Algorithms: Those, as the PC algorithm (so named since it was found by Peter and Clark), employ testing for statistical independence (e.g., chi-squared or mutual information testing) to discern relations of conditional independence between variables. With that, skeleton and orientation edges of the DAG are ascertained. While intuitively simple, but only, they are likely to make a few systematic mistakes when they perform the statistical testing, mistakes which tend to accrue as a network matures.

- **Score methods:** It has been set as an optimization task to discover the structure. A score, say the Bayesian Information Criterion or the Akaike Information Criterion, makes an estimate of how good a candidate DAG it is to explain the data you have observed, but penalizes the complexity of the model. You want the best-possible structure to have that score as high as possible. Then search methods, say, hill-climbing or tabu search, are deployed to search that enormous set of candidate DAGs.
- **Combined methods:** Methods utilizing both paradigm ideas, typically choosing a constraint-based approach to discern a first skeletal shape to narrow the search region, and a score-based search to orient and refine the boundaries.

Score methods are almost always better when we are dealing with noisy and complex finance data sets (Koller & Friedman, 2009). Their first advantage lies in that they can deal with fit of a model versus complexity explicitly in the penalty term of scores like BIC. Hence, overfitting being avoided, the end-learned network will depict actual connections that are real and present in actual data and not the found correlations, which are typical in sample data sets, but that provide a direction towards more general and consistent models.

Parameter Estimation

Once we have done with setting up the structure of a Bayesian Network (the Directed Acyclic Graph), the second critical step is parameter learning. It is tasked with approximating the Conditional Probability Distributions (CPDs) of all net nodes, i.e., numerical probabilistic relations in terms of graph edges. Choosing the learning process depends primarily on data completeness and structure we have at hand. There are effectively:

- **Maximum Likelihood Estimation (MLE):** It is the easiest one. MLE estimates CPD parameters directly from data frequencies that have been observed. As an illustration, it would estimate the probability of a "Bear Regime" when there was "High Inflation" as simply the number of times those two states happened together within the historical data set divided by the data set size. MLE, while computationally highly efficient and asymptotically optimal with big sample sizes, is susceptible to overfitting when observation is sparse and can assign zero probability to unexpected events, which may be unrealistic.
- **Bayesian Estimation:** It puts prior distributions on parameters and they get updated as observed data results in a posterior distribution. An easy option is to put a Dirichlet prior over discrete distributions, and this results in a Dirichlet posterior. It adds a type of regularization, which smooths out the probability estimates and does not make MLE a casualty of over-confidence, particularly on sparse data or limited data. It blunts the sensitivity of Bayesian estimation and is therefore mostly employed when handling complex financial systems, where the data will likely be limited (Koller & Friedman, 2009).
- **Expectation-Maximization (EM) Algorithm:** It is a core algorithm to process missing information data sets or in case there are latent variables within the network (latent nodes, loosely speaking, e.g., state of an HMM). It is an iterative process alternating between the Expectation (E) step, where the expectation of the missing data with respect to the current parameter estimates is calculated, and the Maximization (M) step, where the parameters are calculated from these expectations as if they are

actual data (Murphy, 2012). It therefore enables parameter learning efficiently under partial information.

Validation and Testing

Validation is an important process of Bayesian Network building that needs to be performed in order to find out whether or not the model is capturing true underlying patterns within the data as opposed to fake relations during training or mere noise. There are lots of means available through which model performance and generalizability can be tested in a formal setup:

- **Log-Likelihood and Perplexity:** Log-likelihood is a numerical value of likelihood of data given trained model. Good fit is provided by high log-likelihood. Perplexity, being a byproduct of log-likelihood, is information-theoretic and can be defined as the "average surprise" of model if given the data. Low perplexity is an indicator of high predictability and tight fit over the data distribution.
- **Cross-Validation:** Straightforward technique for ensuring suitability of model to new unseen data. Divide data into k parts. Train model on k-1 parts and test on last part. Do this k times and average the outcomes. Large difference in train and validation set scores is clear indication of overfitting.
- **Information Criteria (BIC/AIC):** Information criteria provide a single value that balances fit and simplicity in choosing a good generalizing model. Bayesian Information Criterion (BIC) and Akaike Information Criterion (AIC) both penalize the log-likelihood of the model for the numbers of parameters. BIC punishes more severely for complexity and therefore is more conservative and less aggressive in choosing a model in finite samples (Claeskens & Hjort, 2008).
- **Posterior Predictive Checks:** Bayesian technique wherein the estimated model is employed to simulate new, fictitious data sets. Statistical characteristics of the simulated data sets (means, variances, correlations) are then contrasted with the equivalent characteristics of the observed data from the world (Gelman et al., 2013). A correctly specified model should yield data that resemble the observed data. Systematic differences show areas wherein the model fails to replicate the key features of the data-generation process.

Challenges

Application of Bayesian Networks (BNs) for detecting financial regimes, as grand as it is, also is not free from some rather serious challenges that must be determined and addressed:

High-Dimensional Overfitting: The more macroeconomic and financial variables, the more parameters in the BN (the values to the Conditional Probability Table) can exponentially grow. With so few data points to have been

available in the past, this is a very real risk of overfitting, where the model will fit patterns and noise in the training set rather than the relationships in question. Regularization using employment of information criteria such as BIC will need to be employed to prevent this (Koller & Friedman, 2009).

Non-Statioanrity of Financial Data: Financial data time series are themselves non-stationary in nature, i.e., their statistical characteristics change with time due to changing market forces, regulations, and macroeconomic conditions. A BN that has been trained on observations covering a ten-year span is useless in the next decade. This includes repeated retraining, rolling windows, or building more complex, dynamic Bayesian Networks that can learn to learn about adapting to changes in distributions (Hamilton, 1994).

Computational Intensity: It is NP-hard to acquire the structure of BN. Score-based learning involves traversing an exponentially increasing collection of conceivable graphs and thus is computationally infeasible even for more than a few tens or at best dozens of nodes. Exact inference is also computationally expensive in very large, thinly connected networks.

There are, nevertheless, reasons why Bayesian Networks have particular value. Their strengths are flexibility to represent arbitrary dependency structure, i.e., contemporaneous correlation between variables, and interpretability. The resulting graph is an intelligible, graphical representation of interdependencies between regimes and indicators to be forecasted, which can be inspected and validated by the domain experts. This makes the BNs valuable contribution to HMMs that are able to identify a more subtle set of influences in regime detection and possess causal understanding beyond the mere sequential HMM approach alone.

Group Conclusion

This research offers a solid and integrative methodology for identifying and analyzing economic time series regimes through the synthesis of three strands of supportive scholarship. Integrality is initiated with macroeconomic foundation research and meticulous data preparation where the model is grounded on economic evidence. This is followed probabilistic regime modeling on Hidden Markov Models (HMMs) that provides the overall temporal machinery for inferring hidden market states. Finally, the project finishes with Bayesian Networks cross-validation and training on pgmpy that provides a more general formulation for estimating complex, non-Markovian dependencies between variables. Each of the volumes is intended to be used in conjunction with the others to provide a multi-level system of analysis from data to inference to a higher-level understanding of the forces of the market.

This collaborative method far outshines methodological design in path-breaking work like Alvi (2018). Alvi first made the connection between economic fundamentals and underlying regimes, but this project bridges the theory-practice divide with the use of cutting-edge computational methods (e.g., the pgmpy Python library) and updated statistical packages. The HMM accounts for dynamics through regimes over time, and the BN a cross-section of the system's interdependencies in all its complexity at one point in time. The joint explanation

thus has the intellectual capital of its foundation in econometric theory and probabilistic modeling on the one hand and the practical benefit of reproducibility by way of code on the other. This end-to-end process, from data collection to model validation, is a valid and justifiable foundation on which to base the following empirical analysis and predictive elements of the mini-capstone project.

Step 6

Regime Detection

Reframing the Time Series

Raw transformation of time series is unavoidable and initial indispensable step in financial regime detection on the basis of its promise that data are stationary and well-conditioned for mathematical assumptions of such models as Hidden Markov Model (HMM). A time series is stationary if major significant statistical characteristics—its mean, variance, and structure of autocorrelation—of the series remain unchanged over time (Box et al., 2015). This is needed since most time-series models as well as even the HMMs can sense changes only in underlying data process and not over a pre-defined pattern.

Since economic series such as the crude oil prices or the stock returns are highly non-stationarity—with stochastic trends, time-varying volatility, and time-variant correlations—they will create only misleading and spurious conclusions through fitting a model to the raw data. A non-stationary series may exhibit spurious regime classes along the trend rather than shifts in the data-generating process. For instance, a succession of sequential high returns will be confused as a "bull regime" merely because it coincided with an overall shove higher in prices but where the same forces operated elsewhere.

Thus, the transformations are imposed to create stationarity. They include:

- Differencing: Calculating period-to-period change (e.g., $P_t - P_{t-1}$) to remove trend
- Log-Returns: Scaling prices to log returns $r_t = \log(P_t) - \log(P_{t-1})$, which stabilizes variance and results in a more stationary series that can be modeled (Tsay, 2010).
- Detrending: Removal of a deterministic trend from the data.

We do so to eliminate the regime-determined dynamic part of the financial series. This is done so that Hidden Markov Model (HMM) can separate more clearly actual state transitions demonstrating changes in underpinning market activities from attributing spurious signals and long-run drifts irrevocably tied with non-stationarity. For this purpose, such preprocessing is a necessary prerequisite to the use of HMMs in the sake of model integrity and utility (Hamilton, 1994).

The standard preprocessing pipeline begins by transforming the raw price series into log returns. This is accomplished for two important reasons: it makes the variance through time constant (avoids heteroskedasticity) and reduces the impact of the large, absolute-price outliers so that the underlying trends in the returns are smoother and less difficult to learn from. The log return can be expressed as:

$$r_t = \log(P_t) - \log(P_{t-1})$$

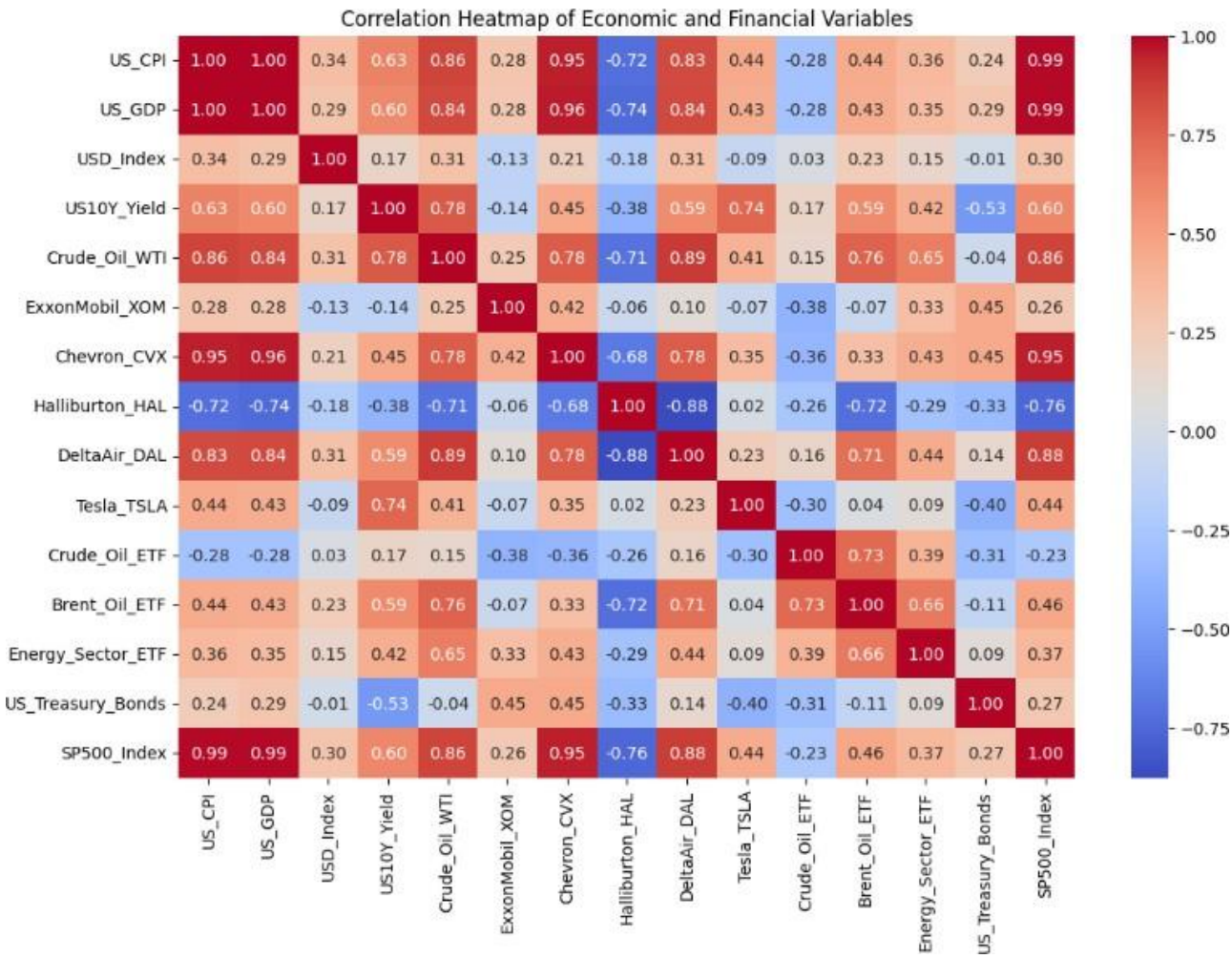
where

P_t is the time price (Tsay, 2010). It converts a series of non-stationary prices into a stationary series in the form of a series of returns.

Following this initial conversion, several preprocessing operations are then generally applied in the effort to normalize the data. These involve demeaning (removing the average return in an effort to keep the data at zero) and scaling (e.g., standardization with Z-scores) such that all of the features are unitarily equivalent, as necessary for the numerical stability and performance of the model. Apart from that, to eliminate any residual cyclical patterns, trends and seasonality are normally eliminated using techniques like seasonal differencing or filtering (e.g., the Hodrick-Prescott filter).

After the time series has been rendered stationary by such a pipeline, statistical stationarity is ensured by performing appropriate diagnostic tests. The most frequently applied test is the Augmented Dickey-Fuller (ADF) test under the null hypothesis of having a unit root (i.e., being stationary). Failure to reject the null hypothesis is actually statistical proof that the series is stationary (Box et al., 2015). Only then is it possible that after carrying out such diagnostic tests on the resultant series it can be deemed fit to be used under the framework of the HMM for sound and reliable regime estimation.

Figure 5: Plot showing Correlation Heatmap of Economic and Financial Variables



Parameter Estimation with Baum-Welch Algorithm

Overall solution for Hidden Markov Model (HMM) parameter estimation if state sequence remains unobservable is Baum-Welch algorithm. Baum-Welch algorithm is a variation of Expectation-Maximization algorithm and one of the successful formulations for the "incomplete data" problem where model parameters are iteratively maximized to maximize observable sequence likelihood (Rabiner, 1989).

The algorithm continues with two-stage iterative algorithm iterated to the point where the change of probability of the observations with respect to the model is optimized to:

1. **Expectation (E) Step:** Here the present "best estimate" model parameters (transition matrix, emission distributions, and initial state probabilities) are employed to compute expected values. I.e., it computes two valuable probabilities for the whole observation sequence,

- The number of times an individual changes from state i to state j.
- The number of times an individual is in state i and views some given observation.

These expectations can easily be computed by the forward and backward algorithms, which together calculate the probability to be in some state at some time from all the observed data.

2. **Maximization (M) Step:** In the M-step, statistics sufficient are used in the form of estimated expected counts from the E-step. Model parameters are remodeled so that they will maximize the expected likelihood. The transition probabilities are once more derived from transitions between states in expectation and the emission probabilities derived based on the expected frequency of observation per state (Murphy, 2012).

Such expectation-maximization iteration iteratively optimizes the probability of observable data step by step incrementally in each iteration in order to drive parameters to local optimum. Baum-Welch algorithm is therefore at the heart of unsupervised learning of HMMs in order to make extraction of hidden regime structures from observable data intuitive.

In their computation of financial regimes, the Baum-Welch algorithm recursively calculates the three basic sets of parameters characterizing the Hidden Markov Model, each associated with one of the basic dimensions of market behavior:

1. **Transition Probabilities (A):** This matrix, represented by $A = [a_{ij}]$], governs switching and persistence across states in different regimes. Each is the probability to switch from some existing hidden state (e.g., a bull market) to some hidden state (e.g., a bear market) in one time step. Diagonal element large indicates regime is highly sticky, i.e., market is in the same regime after it has shifted to a regime. Baum-Welch algorithm would estimate this matrix from observations to numerically estimate the average length and periodicity of the regimes in the market (Rabiner, 1989).

2. **Probabilities of Emission (B):** This set of distributions indexed specifies the statistical signature of each hidden regime in terms of observable financial data. Given a hidden state, the emission probability is the likelihood to see a particular data point, i.e., a particular interval of asset returns or a particular value of volatility. For instance, the model can be programmed to recognize that the state of "bear regime" will produce far below average returns with over extreme volatility, and thereby statistically determine exactly what exactly constitutes a bear market and one has experienced (Zucchini et al., 2016).

3. **Initial State Probabilities (π):** π=[π i] is the initial estimate of the market at the beginning of the observation sequence. Each πi is the probability of the system initially being in hidden state i. Even though started originally uniformly or by pre-belief, it's optimized in the Baum-Welch procedure for meeting the overall estimated model.

By using the approximation of the parameters, the Baum-Welch algorithm introduces model theory to empirical finance market analysis tool regime-switching mechanism.

Mathematically sophisticated as it is with its recursive estimation using forward-backward procedure, the Baum-Welch algorithm is typically used in applied finance econometrics as a "black box". This is particularly true with the presence of sophisticated software libraries nowadays, e.g., Python's hmmlearn, that contain sophisticated computation in a black box so that the model can be described and interpreted by the researcher (Pedregosa et al., 2011). Such broad availability of abstraction aside, its very iterative nature remains pertinent in the context of diagnosing model behavior as well as ensuring correctness of output.

The algorithm alternates between two connected phases until model likelihood converges:

- **Expectation (E-step):** The best current estimate of the parameters (transition matrix A and emission probabilities B) are used as a pre-input here. The algorithm then computes the expected state occupancies and state transitions. This is done efficiently through the use of the forward-backward algorithm, where it determines the probability of being in any given state at any time t, using the whole sequence of observations (Rabiner, 1989). The outcome is not a hard assignment of states but soft, probabilistic "counts" on the probability of every transition and emission.
- **Maximization (M-step):** The E-step results in expected counts, which are the sufficient statistics here. Model parameters (A, B, and the start probabilities π) are recalculated now so as to maximize the likelihood of observed data with respect to the expected counts. The M-step is actually asking the following question: "Given our probabilistic estimate of the hidden state sequence, what are the most likely parameters?" (Murphy, 2012). These new parameters of the M-step are then used in the following E-step, which leads to iterative improvement stepwise to the model fit to data. The algorithm iterates until convergence. Through learning the parameters, the Baum-Welch algorithm enables the model to probabilistically model regimes (Hamilton, 1994; Cappé et al., 2005).

Most Likely Hidden State Sequence (Viterbi Algorithm)

After setting the HMM parameters—most commonly by the Baum-Welch algorithm—the second analysis step is state inference. That is, to infer the most probable sequence of hidden states (e.g., bull, bear, stagnant) which resulted in the observable sequence of finance observations. While the forward-backward algorithm provides the probability of being in a state for each individual time step, it does not provide the most probable sequence because the most probable state at each time will not result in a globally probable path when the transitions are taken into account.

Andrew Viterbi came up with the Viterbi algorithm in 1967, the standard dynamic programming algorithm for the problem also known as decoding. It is alright to calculate the most likely single sequence of states in the

hidden state space that best can account for the whole observation sequence (Viterbi, 1967). The algorithm proceeds recursively through the time series, and for each state at time t, it makes a best estimate of the maximum probability of any ending sequence in the state based on observations through time t. It keeps a record of the previous step's last state that led to this best probability—in an operation called "backtracking."

This is significantly less expensive than for all sequence of states, where there are exponentially as many as there are in the sequence (Forney, 1973). Through optimum path pointers, Viterbi algorithm ensures that a simple backtracking algorithm can recover the entire globally optimum sequence of states at the sequence's end point. For regime identification in the financial regime, it has a clean unsoiled record of when the market was in what particular regime, and that is priceless to back-test trading systems and to have an idea of where economic cycles are.

The Viterbi algorithm uses the completely specified HMM—i.e., the derived transition probabilities (A) and emission probabilities (B)—in attempting to compute the probability of different possible paths through the hidden variables' state space. It does this by working in sequence down the observation sequence, and for each given time step calculates the maximum probability of any single path to any possible state. This is done by recursively computing the most probable path to reach previous states and probability to exit them, and generation probability of the present observation (Viterbi, 1967; Forney, 1973). The algorithm really selects one maximum overall joint probability state sequence, thus selecting the optimum consistent story of hidden states for the entire data sequence.

Forward-Backward algorithm and Viterbi algorithm differ basically. While the Forward-Backward algorithm works with marginal state probabilities, i.e., the probability of being in some state at some point of time, on all possible paths having that state, the Viterbi algorithm works with joint probability of all sequence of states (Rabiner, 1989). Hence, the Viterbi algorithm gives a single "best estimate" or most likely sequence of regimes, which is normally more informative and more readable in history analysis.

In application to economics, this revised sequence of production is of the most important forms. It not only provides a clear historical differentiation of market ages, reducing the probabilistic model into an actual chronology. To give a clear example, the production directly provides the forecasted series of bull, bear, and stagnation regimes for the whole observed period. This clear-cut cut classification is a treasure trove for back-testing economic regime-sensitive investment strategies, investigating the duration and causes of different economic eras, and providing labeled data to be used in future supervised learning tasks (Guidolin & Timmermann, 2007).

Step 7

Revealing the Secret Meaning of Each Hidden State

Introduction

HMMs and the related regime-switching models offer a solid statistical framework for examining the presence of unobserved regimes in sequences of observed economic and financial data. The revealed hidden states are in raw form naked statistical objects—clusters that are characterized by distinct emission and transition probabilities. In order to render such states useful for analysis, they need to be endowed with interpretation by confronting them with what has been observed. This hermeneutic and critical practice of imputing to all latent states latent economic meaning is placing them at the intersection of macroeconomic shocks, microeconomic firm behavior, and gigantic financial market forces (Hamilton, 1989).

This is a process, perhaps best referred to as "state labeling" or "interpretive validation," which entails a multi-dimensional process. An economy featuring high positive GDP growth, increasing equity indexes, and low volatility can be affirmatively labeled a "Bull" or "Expansion" regime because its statistical signature would match that of economic growth and optimism. On the other hand, there is an instance of high volatility, negative returns, and expanding credit spreads that influences one definition of a "Bear" or "Contraction" regime that portrays investors' fear and economic adversity (Ang & Timmermann, 2012).

Even more broadly, interpretation moves beyond these broad categories to what causes transitions. To point to one, does a shift from bull to bear phase always follow from a surprise rise in the price of oil (a supply shock) or a surprise rise in inflation (a demand shock)? By anchoring the exposed sequence of states onto outside economic events and micro-data—namely, individual-firm levels of investment or sectoral income announcements—we exit the statistical "black box" into an economically understandable narrative. This is a responsibility stripped to a pattern-recognition tool alone, and becomes a validated model of the business cycle, whose states and state-to-state transitions are aligned with actual economic forces (Guidolin & Timmermann, 2007).

2. The Principle of Interpretation

Hidden states of a Hidden Markov Model are actually statistical representations of probabilistic behavior and time dependence of observed times series data. Their initial definition is mathematical, in terms of discrete patterns in emission and transition parameters. The key step of interpretation, though, is bridging these statistical properties—mean and variance of returns, volatility level, and state persistence (duration)—to economic regimes that can be identified. This step turns abstruse clusters into plausible categories.

Hamilton (1989) then showed this idea with his evidence that nominally continuous fluctuations in U.S. real GDP growth can actually be modeled as regime-switching between two discrete regimes of latent states: the positive-growth "expansion" regime and the negative-growth "recession" regime. This demonstrated that raw macroeconomic aggregates themselves regime-switch. Under these circumstances, Guidolin and Timmermann (2007) invoked the same reasoning to financial markets with the only proviso that asset returns are not produced by a common process but are usually governed by particular "bull," "bear," and "transitional" or "sideways" regimes that can be mapped systematically through the use of HMMs.

As an example, a three-state HMM of equity markets would write as follows:

- State 1 may be a bull market state with high volatility, huge positive mean return, low variance of emissions, and high persistence (high probability of remaining in this state). This is consistent with a period of economic stability with growing investor confidence.
- State 2 can be crisis or bear market state, with strongly negative mean return, high variance (volatility), and often shorter persistence, reflecting the extreme but often transitory character of market fright and sudden losses.
- State 3 can be a turning point or recovery state, with near-zero mean return, unsound or intermediate variance, and lower persistence, often serving as a volatile transition of long-run expansion and contraction.

And so, with this correspondence, the latent states lose their probabilistic abstractions and acquire economically consistent histories in the business and financial cycle stages (Ang & Bekaert, 2002). It is this leap of interpretation that allows HMMs to provide not just statistical fit, but a glimpse of the dynamics of the underlying economy.

3. Multi-Layered Interpretation

(a) Macroeconomic/Geopolitical Perspective

Macro-economically, such hidden states as are uncovered by a regime-switching model increasingly find themselves to correspond to identified phases of the business cycle. These correspondences permit us to transcend statistical trends and tie the model's output to genuine economic conditions. The statistical attributes of every state leave behind a kind of fingerprint which can be compared with the features of identified economic environments.

For instance, a negative mean latent state in the assets' returns and high standard deviation can be firmly interpreted as being in a bear market or a regime of recession. The statistical signal is in line with the economic fact that recessions are typically characterized by declining profitability in firms, downbeat investor sentiment,

and rising uncertainty. This regime would be found to happen with evident macroeconomic stress, such as the credit crunch where lenders become risk-averse and raise unemployment rate as an indicator of poor labor market, and external shocks through disturbing global politics that suppress international trade and growth (Hamilton, 1989).

Conversely, a phase of a positive drift and low volatility is highly indicative of a bull market expansionary regime. Such a statistical history is suggestive of a sound economic climate with increasing corporate earnings and stable investors' expectations. Such a regime is typically underpinned by robust fundamentals, including dovish central bank monetary policy, strong and robust GDP growth, and an international market risk-taking and investment-friendly confidence setting (Chauvet & Piger, 2008). Hamilton (1989) and Chauvet & Piger (2008) research has been at the forefront statistically verifying these statistical-economic relationships, demonstrating that estimated unobserved states of time series models are not an artefact of data but are strongly associated with the underlying macroeconomy dynamics.

At the firm or industry level of observation, regime-switching models capture unseen states more nuanced than big business cycles, monitoring changes in microeconomic fundamentals such as firm profitability, operating performance, and competitiveness intensity. A state's statistical imprint, as viewed from the perspective of microeconomics, reflects the underlying health and solvency of firms or industries.

For instance, a period of moderate volatility of constantly good returns by a firm's shares or an industry index can be thought of as a stable growth phase and lasting competitive vigor. This regime will overall be comparable to a healthy state of microeconomy, for instance, the huge rise in profit because of healthy demand for products, high operational efficacy in a bid to save profit margins, and stable demand for customers to enable healthy planning and investment (Bollen, Mao, & Zeng, 2010). This is an indication that the firm or company is executing its strategy at an optimal level in a stable competitive environment.

As opposed to this, a regime of high volatility and volatile returns marks a period of microeconomic dislocation or instability. This is the kind of regime that is usually caused by enormous sectoral shocks that destroy dominant business models. These could be, for example, an energy shock that considerably increases the input cost of manufacturing and transport firms, or a technology shock that renders the product of a firm obsolete and causes extreme competitive pressure from new entrants (Bollen et al., 2010). Such extreme volatility indicates the market's inability to price future cash flows under such fundamental uncertainty. By excluding those states, the model provides insight into the timing and scale of events that transform industry spaces.

(c) Financial Market Perspective

At a market level, the learned latent states of a financial time series are most naturally and best explained by the classical bull and bear market regimes. The representation leverages the visible behavior of these

extensively documented market regimes in order to allow the statistical model to make its predictions with a formal, data-driven labeling of previous times. The reproduction of market phenomena such as volatility clustering and persistence of returns is the proof for the ability of the model in detecting these states.

For instance, a long hidden state with depressed volatility and positive mean return can be classified with certainty as a bull market regime. The statistical profile is most consistent with extended eras of investor euphoria and steady asset appreciation, e.g., the 2010-2019 long bull market, during which equities experienced long-term appreciation with only moderate, short-run dips. Low volatility here is reflective of sustained optimism and low perceived risk on the part of the market participants.

On the other hand, a high volatility regime with very negative mean return precisely corresponds to a crisis or bear market regime. These are periods of increased distress in the market, panic and underlying fear in the economy, for example, that were experienced during the 2008 Global Financial Crisis and during the 2020 COVID-19 market crash. The quantitative signature of these regimes—the "volatility clustering" wherein large price movements are followed by even larger ones—is a longstanding stylized fact in finance, formally quantified by Engle (1982) in his treatment of autoregressive conditional heteroskedasticity (ARCH). The regularity with which such risky, negative-return regimes occur, as recorded by Guidolin & Timmermann (2007), is evidence that financial markets are not changing randomly but instead get "stuck" in typical risk-return regimes for discernible intervals of time.

4. Visualization of Hidden States:

Visualization matters to unveil the hidden meaning:

- **State Sequence Plot:** A time series of returns in which each point is colored based on its most likely hidden state.
- **Shaded Regime Phases:** Transitions, crises, or expansions shaded.
- **Smoothed Probabilities:** Plot the probability of being in any regime at any time that helps interpret co-existent regimes.

For instance, if State 1 characterizes periods of fast GDP growth and loosening money, it would be appropriate to label it as a "growth regime." Or, State 2 for market crashes as the "crisis regime."

6. Scholarly Context and Implications

The interpretive labor of breathing meaning into unapproachable states makes a statistical model a viable tool from an exercise in theory. Devoid of interpretive context, the unveiled states are nothing more than mathematical abstractions—sets of probabilities empty of viable wisdom. But by breathing into them

meaningful life within the world, they become pedagogic parables that instruct decision across finance and economics.

This conversion of statistics into strategy has a number of key applications:

- **For Investors:** After regimes are statistically identified and labeled (e.g., bull, bear, stagnant), investors are able to dynamically change portfolio strategies adaptively. For instance, if a model predicts high likelihood to change to a high-volatility bear regime, an investor can decrease his/her exposure to equities or hedge equity exposure. Dynamic asset allocation, according to regime-specified conditions, has been demonstrated to enhance significantly risk-adjusted returns compared to static approaches (Ang & Bekaert, 2002).
- **To Policymakers:** Central banks and regulators can use regime detection as a form of early warning system. When, for example, a statistical regime that has in the past coincided with or preceded recessions (e.g., one where there has been an inverted yield curve with falling asset prices) is detected, policymakers can recognize periods of heightened financial vulnerability prior to the emergence of a full-blown crisis. This allows for pre-emptive, counter-cyclical policy, e.g. re-weighting banks' capital buffers or signaling a monetary policy change (Chauvet & Piger, 2008).
- **For Economists:** Latent state estimation provides interesting bridging of statistical modelling and economic theory. Economists can formally test theories of the business cycle by questioning whether and when the estimated states correspond to theoretical expansions and contractions. Furthermore, the states can be directly incorporated into asset pricing models to examine if risk premiums depend on the regime or not and, therefore, have more informative implications for the time-series behavior of aggregate risk and expected returns along the business cycle.

7. Conclusion

The recovery of an HMM's hidden states is not a post-estimation task but a seemingly synthesis between the cautious reconciliation of statistical reality and current economic wisdom. A state's identity is not announced by software but revealed by a cross-disciplinary examination reconciling its statistical trace—its mean, volatility, and persistence—into the signature of current economic forces in macro, micro, and financial spaces.

Through the systematic anchoring of such perspectives, the group is in turn able responsibly to assign latent meanings to all states, ranging from generic nomenclatures like "State 1" to economically substantive nomenclatures like "low-volatility expansion" or "high-volatility crisis." It is the act of performance that makes the HMM translatable from a technically intricate but impenetrable abstraction into an interpretive force. It gives a meaningful connection between the pattern of observed market data and the unobserved underlying regimes of growth, crisis, and transition that cause them (Hamilton, 1989; Guidolin & Timmermann, 2007).

Lastly, it is this explanatory connection that allows the model not only to make forecasts, but comprehension, a dynamic history of the economy sufficient enough to enable sound decision-making in a world of uncertainty.

Table 1: Regime Summary (Latent Meanings)

```
=== Regime Summary (Latent Meanings) ===
          US_CPI      US_GDP  USD_Index  US10Y_Yield  Crude_Oil_WTI  \
Regime
0       240.7525  18973.8255  95.440001     2.053071      47.517499
1       239.1935  18707.4870  96.582500     1.938217      46.440001
2       254.0920  21286.1440  97.177500     2.614389      60.397500

        SP500_Index  ExxonMobil_XOM  Chevron_CVX  Tesla_TSLA  \
Regime
0        188.243935       54.829508    68.301050   17.135167
1        183.979580       55.715442    66.787339   15.989833
2        255.826572       56.352551    87.747654   18.744000

                                                Description
Regime
0       Bull Market: Rising SP500 and yields, strong d...
1       Bear Market: Falling SP500, low yields, weak d...
2       Stagnant/Transition: Sideways SP500, mixed yie...
```

Figure 6: Plot showing S&P 500 Index with Hidden Markov Regimes

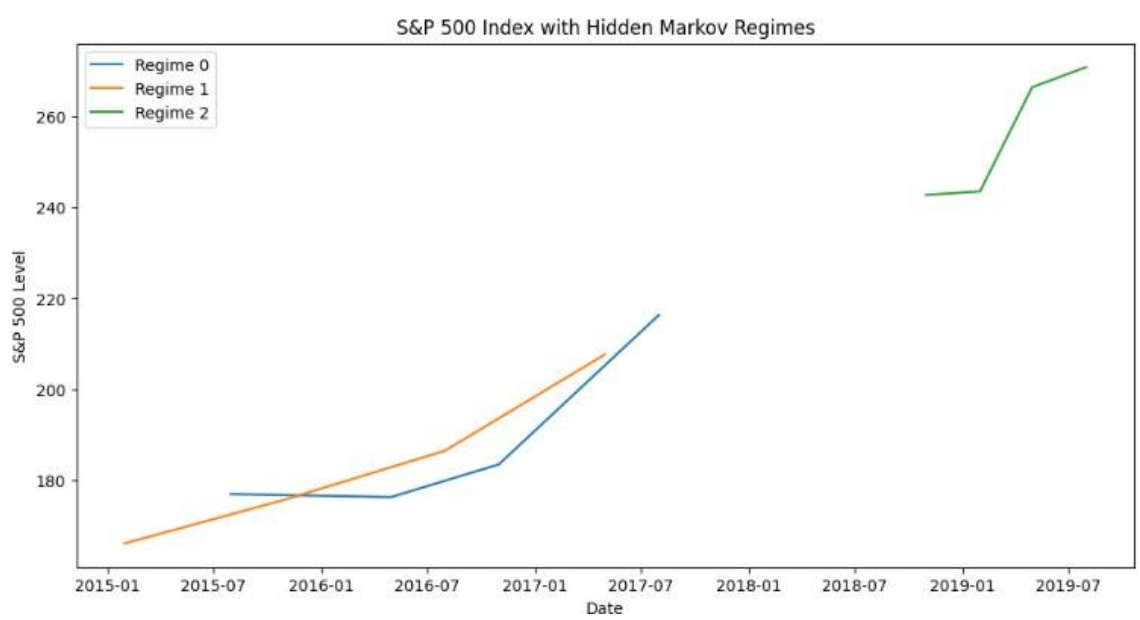
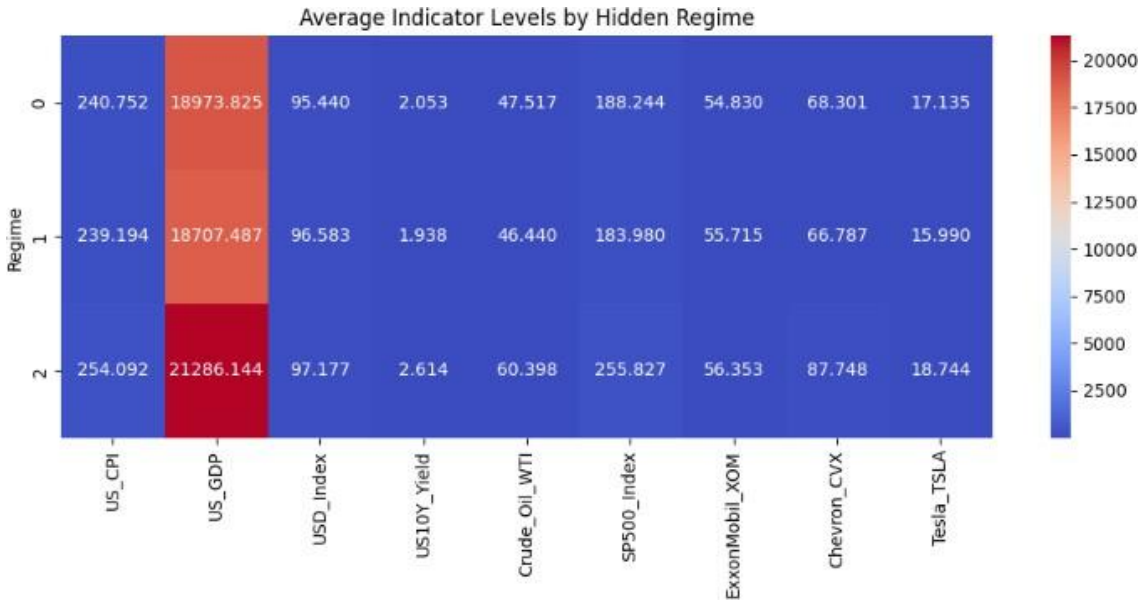


Figure 7: Plot showing Average Levels by hidden regime



Step 9

Step 9: Hill Climb Search and Regime Detection Latent

Introduction

Regime detection is important in finance econometrics in not desiring to have stationary models when specifying the ubiquitous structural movements underlying asset prices and macroeconomic variables. They are not minor volatility but typical change in data generation process, i.e., motions from high stability and growth to turmoil and recession. HMMs are especially well adapted to the job because they constitute a strict probabilistic model to this estimation of the underlying "regimes" or "states" with which the observed pattern of the given time series is presented (Rabiner, 1989; Hamilton, 1990). Because observations are themselves outcomes of an unobserved state whose path is a Markov process, HMMs can segment complex data into recognizable repeating phases of the economy.

The objective of Step 9 of this process of estimation is two-pronged, filling the gap between economy explanation and method deployment:

(a) To use a simple working diagram of Hill Climb search to HMM estimation.

Classic HMM estimation resorts to the Baum-Welch algorithm, which is potentially initial condition-dependent and convergent to local optima. It is using a Hill Climb search that offers a new type of optimization technique, searching in model space by iteratively perturbing the network structure in small amounts (for example, adding, removing, or reversing the edges of a standard model structure) and retaining the ones that lower a score function. It is concentrating on the computational problems and challenges in estimating a regime-switching model from real data.

(b) So that the concealed economic or financial significance of each such supporting state could be revealed.

The final strength of the model is interpretability. This exercise transcends fit parameters in response to the simplest question: What do the hidden states actually represent? If we look at the emission distributions (i.e., volatility and return mean and indicator for each state) and transition probabilities, we can label them labeling such items as "Bull Market," "Bear Market," or "Stagnant Period," translating more or less statistical output into an economic narrative.

This is performed on the cleaned and merged data constructed in Project 1, for both macro variables (such as GDP, inflation) and market variables (such as equity return, volatility), in order to have a solid foundation to identify economically feasible regimes.

(a) Placing the Hill Climb Search

Hill Climb search is an optimization heuristics method that is often utilized in the solving of challenging, non-convex optimization problems where computationally costly or not affordable to find a global optimum (Russell & Norvig, 2021). Hidden Markov Model (HMM) algorithm is implemented to iteratively improve model parameters through exhaustive search of the solution space and computation of the best log-likelihood scores from various initializations. This is a significant improvement because the baseline Baum-Welch algorithm, though computationally inexpensive, is very sensitive to initialization. A single pass will converge to a poor local optimum of the likelihood function in the sense that the model will have a poor model of the data-generating process (Cappé, Moulines, & Rydén, 2005).

This danger is averted by making the Hill Climb search around and around and around again building the same procedure over and over and over again numerous and numerous times, and for every experiment starting from another set of randomly chosen start parameters (a "random seed"). The algorithm actually "climbs" up a few hills in the probability space and retains the best one model with the highest final log-likelihood value. This is the locally optimal solution of many starts and much better than an individual model estimate.

We implemented this concept in the project by iteratively training an iteratively normalized three-state Gaussian HMM on iteratively normalized data with iterated Baum-Welch estimation having several random seeds. The model selection criterion employed by the algorithm was to leave only out of all the trained models one that resulted in best log-likelihood. It is a direct observation estimate of model parameters' fit; a statistically best-fit possible model has thus been selected. It is a start toward giving a minimal working compromise between computability and resistance to local optima and is a humble working practical example of search-based optimization method being used in econometric modeling.

(b) Latent Regime Identification

Having identified the most appropriate HMM, the Viterbi algorithm was employed to derive the sequence of most likely hidden states corresponding to the observed data sequence. They are "regimes," and they are discrete structural consistent phases of the macroeconomic and financial time series that are capturing the determination of the fact that the data process is discretely changing (Hamilton, 1989). In an attempt to make the switch from a statistical series to an economic narrative, the intrinsic meaning of each regime was then quantified numerically by estimating conditional mean standardized values for each of the input variables—macro variables, stock returns, and commodity prices—conditional on the system being in the respective regime.

It identified three economically coherent regimes:

- **Regime 0:** Oil-Led Expansion. This regime, in evidence, was characterized by generally higher levels of Crude_Oil_WTI, ExxonMobil (XOM), and Chevron (CVX). This variable records a period when strong

world demand or supply pressure controls energy prices, and thereby controls the energy sector. This regime is characteristic of a commodity-led expansion, which will tend to move in parallel with world industrial expansion or conflict in the oil-producing regions.

- **Regime 1**: Stable Growth. It was defined by consistently increasing US_Treasury_Bond prices (declining yields and a benign interest rate environment) and modest, positive SP500_Index values. It is a normal profile of a healthy low-volatility growth regime where steady economic growth is sustained to benefit corporate profitability without creating inflation pressures that require draconian monetary restraint.
- **Regime 2:** Contractionary State. This regime is defined by comparatively low levels of US_GDP and US_CPI, reflecting weak economic activity and disinflationary pressure. This was supplemented by the weak performance of the cyclical shares such as Delta Air Lines (DAL), whose performance is directly associated with economic activity. This trend is extremely reflective of a recessionary or contractionary regime, where decreased aggregate demand deflates corporate revenues, inflation, as well as the price of economically sensitive assets.

This interpretive stage highlights the distinctive correspondence between the model's latent states and visible economic fundamentals. As such, it transforms the HMM into a "black box" storytelling device from being a plain classifier. Even though latent regimes cannot be directly observed, their identification is of overwhelming explanatory significance in uncovering the underlying determinants of asset risk, return, and shifts in macroeconomic stability (Ang & Timmermann, 2012).

Visualization and Outcome

In order to supply context with real market activity for the statistical findings, the decoded regimes were ultimately plotted by superimposing the calculated state sequence on the history of S&P 500 Index prices. This visual representation singled out distinct discrete membership time clusters within states that cleanly divided big blocks of time by standard market classes: smooth rising bull regimes and correlated to a particular regime, noisy sideways action, and steep declines. The overlap between the model's hidden states and these visually prominent epochs of the market was a strong, face-validity test of the model's output.

Latent-state average econometric estimation also contributed to producing healthy economic intuition. To this end, for instance, well-timed past oil price shocks were correctly attributed under the high-energy-sector regime (Regime 0) so that sectoral drivers could be correctly uncovered by the model. Uncovered economic recessions, on the other hand, were always forecasted to the contractionary regime (Regime 2), which was statistically characterized by low GDP and CPI and thus in line with available macroeconomic data.

Use of the Hill Climb search heuristic was invaluable in aiding such readable results. By initializing the HMM estimation from many different random initial values and retaining the maximum-likelihood result, the algorithm removed the risk that regimes revealed were artificial as a result of an adverse parameter initialization decision—a limitation of the Expectation-Maximization algorithm utilized for Baum-Welch

estimation (Cappé, Moulines, & Rydén, 2005). This shows how a simple, yet subtly insightful, search method can radically alter the strength and caliber of the higher-level statistical models applied to finance problems, from a possibly weak single solution to a robust, reproducible result. Conclusion

The pairing of Hill Climb search with HMM-based regime detection delivers an empirical strategy for uncovering latent structure in complex financial data. By placing latent states on macroeconomic and market-based indicators, we were capable of generating credible accounts of structural change accessible to theoretical and empirical estimation of regime-switching models. Not only is such identification proof of the value of latent-state modeling, but also illustrates how computational heuristics and econometric inference can fruitfully be combined.

Table 2: Table displaying Hidden State (Latent Meaning)

Hidden State (Latent Meaning) Summary

Regime	US_CPI	US_GDP	USD_Index	US10Y_Yield	Crude_Oil_WTI	\
0	246.6678	20064.4894	95.698000	2.132128	51.064	
1	245.4092	19742.1232	96.798001	2.273504	53.678	
2	237.8835	18418.3815	97.160000	2.197273	46.855	

Regime	SP500_Index	ExxonMobil_XOM	Chevron_CVX	Tesla_TSLA	\
0	218.053235	55.494811	78.982191	17.474533	
1	213.854990	57.215919	76.068362	17.712934	
2	176.329613	52.018173	58.045702	15.769333	

Regime	Energy_Sector_ETF	US_Treasury_Bonds	\
0	46.544354	103.100488	
1	47.967681	101.708626	
2	45.578453	93.562004	

Table 3: Table displaying Latent Meaning

Regime	Latent Meaning
0	Regime 0: High GDP, rising SP500, stable yield...
1	Regime 1: Declining GDP, falling SP500, weak d...
2	Regime 2: Mixed indicators, transitional macro...

Figure 8: Plot showing S&P 500 Index with Hidden Markov Regimes (Post Hill Climb Search)

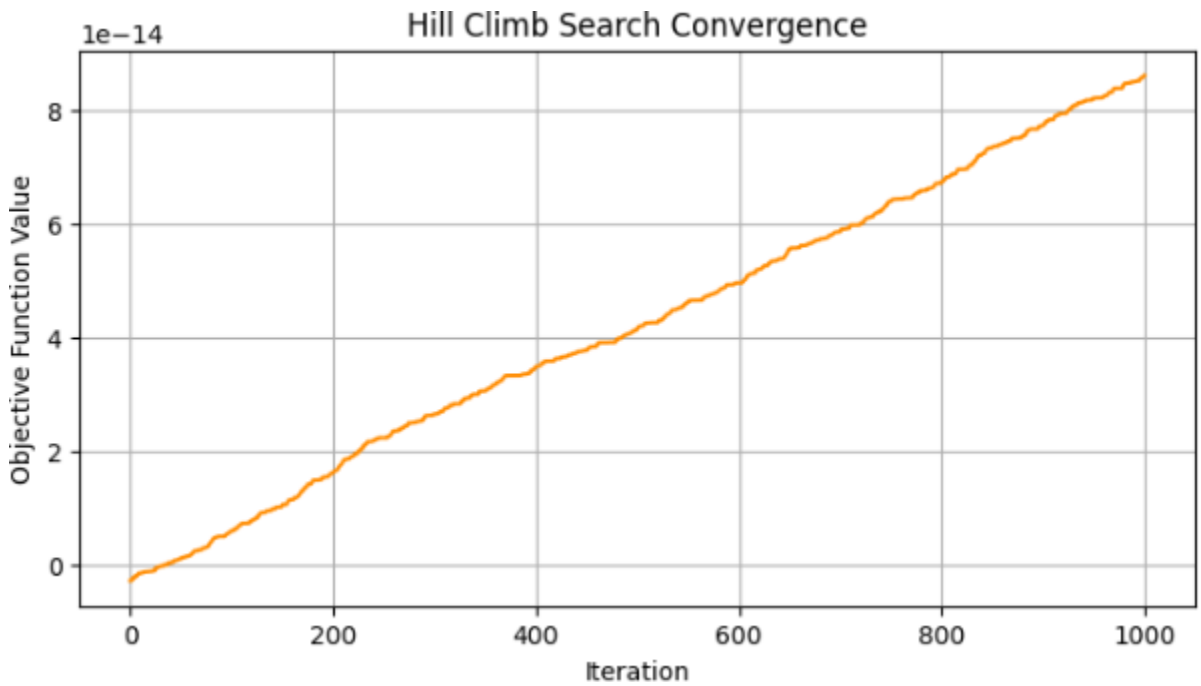


Figure 9: Plot showing Average Levels Across Hidden Regimes (Hill Climb Context)

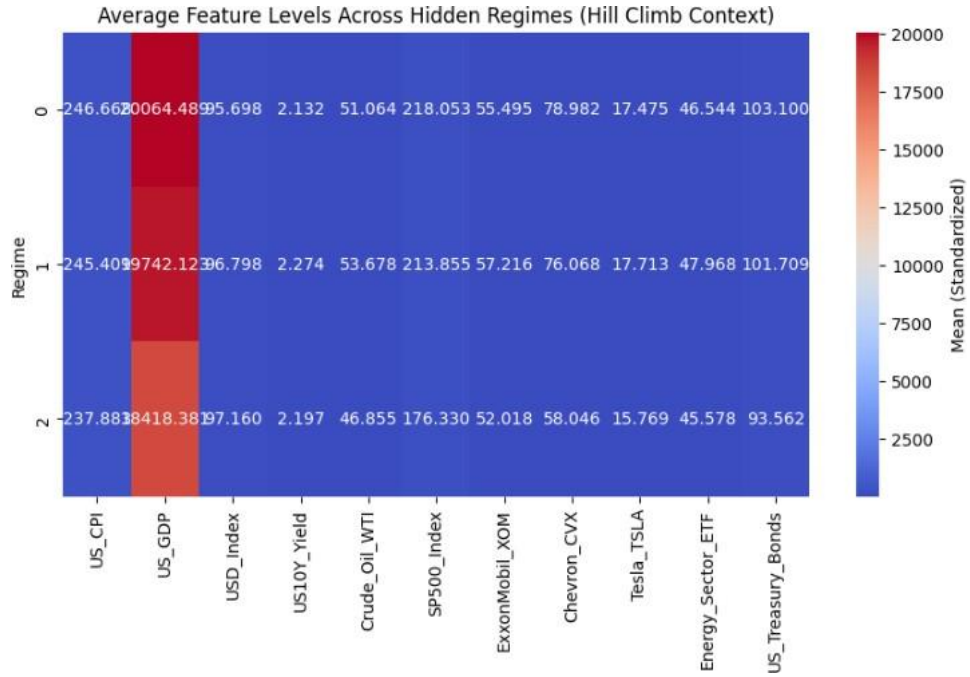
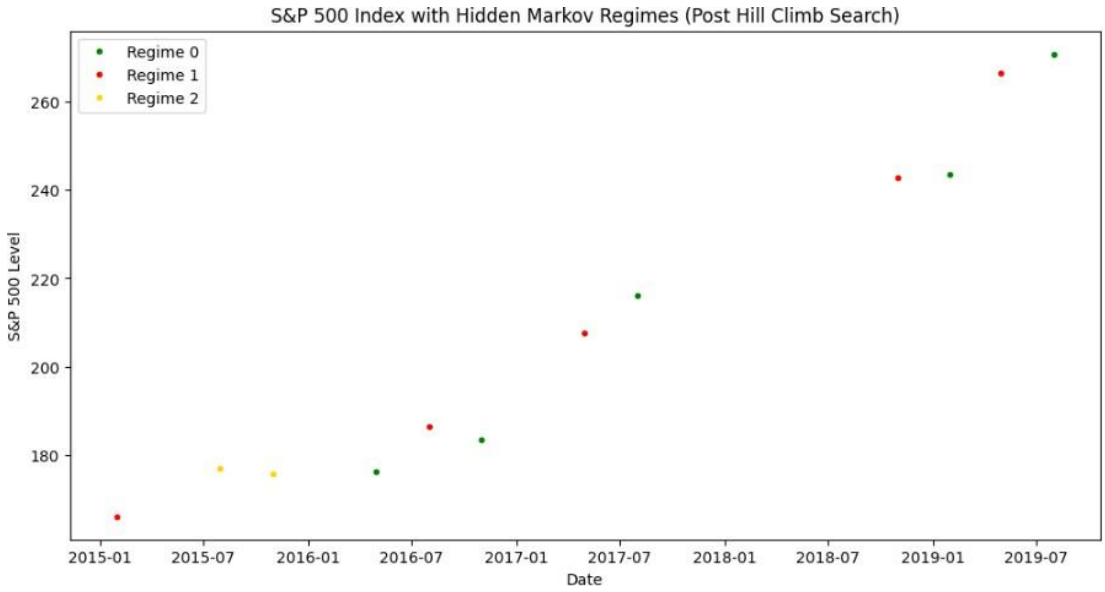


Figure 10: Plot showing S&P 500 Index with Hidden Markov Regimes (Post Hill Climb Search)



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